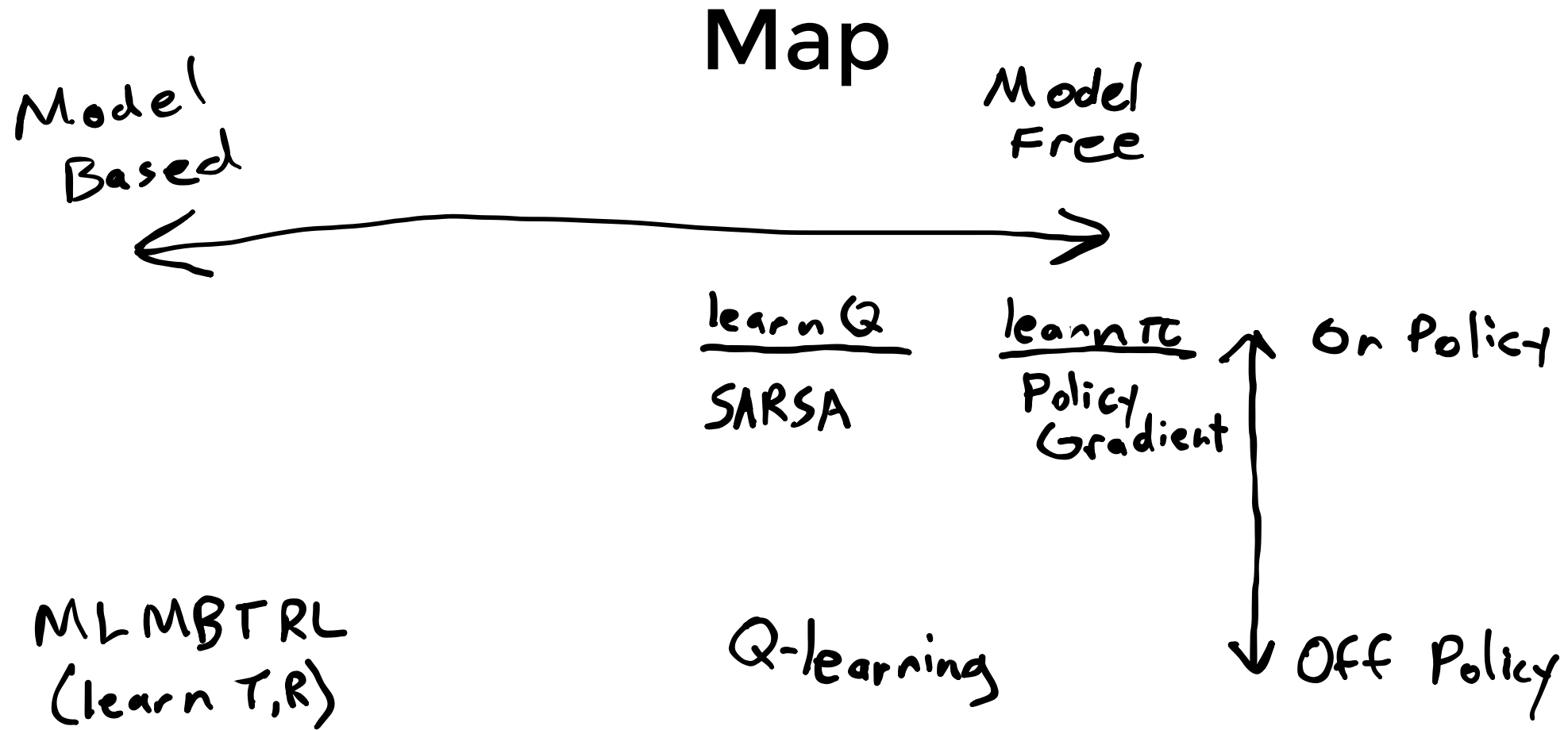


**Actor-Critic**  
**Reward Shaping**  
**Advanced Exploration**  
**Entropy Regularization**



Challenges:

1. Exploration vs Exploitation ←
2. Credit Assignment
3. Generalization

1. Actor-Critic
2. Reward Shaping
3. Advanced Exploration
4. Entropy Regularization
5. Wisdom

# 1. Actor-Critic

# 1. Actor-Critic

Can we combine value-based and policy-based methods?



# 1. Actor-Critic

$$\nabla U(\theta) = E_{\tau} \left[ \sum_{k=0}^d \nabla_{\theta} \log \pi_{\theta}(a_k \mid s_k) \gamma^k (r_{k,\text{to-go}} - r_{\text{base}}(s_k)) \right]$$

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$Q(s,a)$        $V(s)$



# 1. Actor-Critic

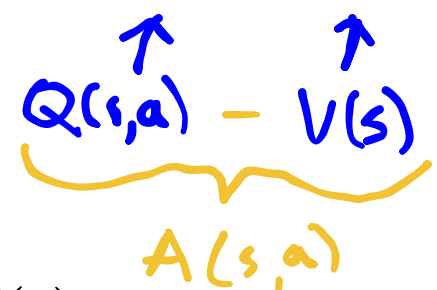
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$\uparrow$   $\uparrow$   
 $Q(s,a)$   $V(s)$

Advantage Function:  $A(s, a) = Q(s, a) - V(s)$

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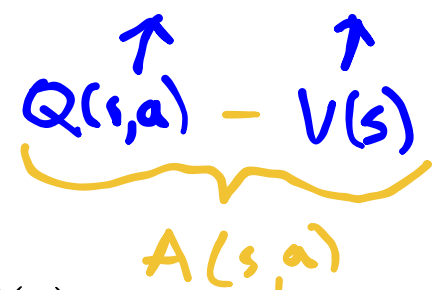
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 $A(s,a)$

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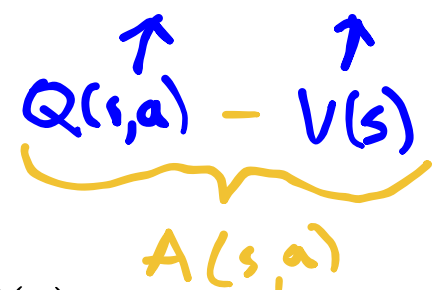
  
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- Actor:  $\pi_{\theta}$

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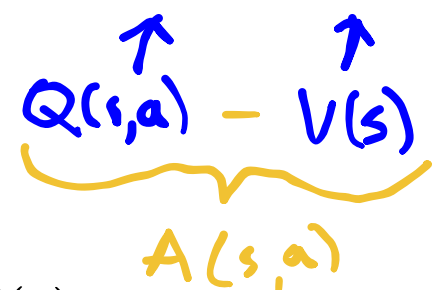
  
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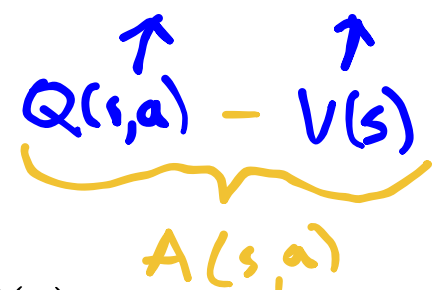
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Alternate between training Actor and Critic

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Alternate between training Actor and Critic

Problem: Instability



# Actor-Critic

ignore baseline  
- state and action

Which should we learn?  $A$ ,  $Q$ , or  $V$ ?

+ Use it directly  
- state and action

+ only function of state  
- doesn't give feedback about action directly

# Actor-Critic

Which should we learn?  $A$ ,  $Q$ , or  $V$ ?

$$\nabla U(\theta) = E_{\tau} \left[ \sum_{k=0}^d \nabla_{\theta} \log \pi_{\theta}(a_k \mid s_k) \gamma^k (r_k + \gamma V_{\phi}(s_{k+1}) - V_{\phi}(s_k)) \right]$$

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↙ baseline

# Actor-Critic

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↑ estimate with  
reward to go  
from sims

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# Generalized Advantage Estimation

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$$A(s_k, a_k) \approx \underbrace{\left( \sum_{t=k}^{d-1} \gamma^{t-k} r_t \right)}_{\text{Use rollout for } d \text{ steps}} + \gamma^{d-k} r_d + \gamma V_\phi(s_{d+1}) - V_\phi(s_d)$$

Use critic

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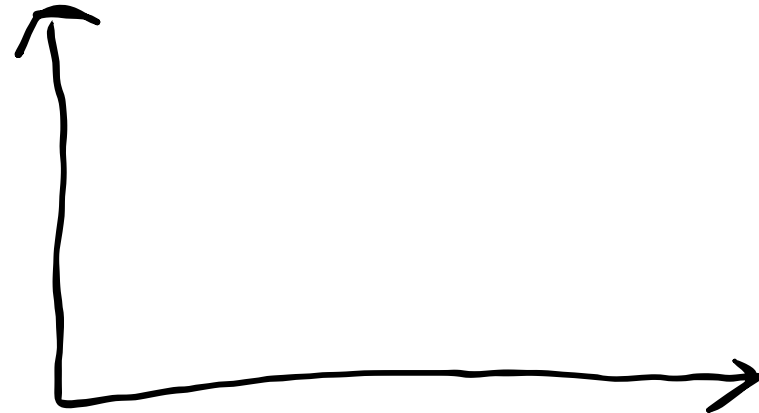
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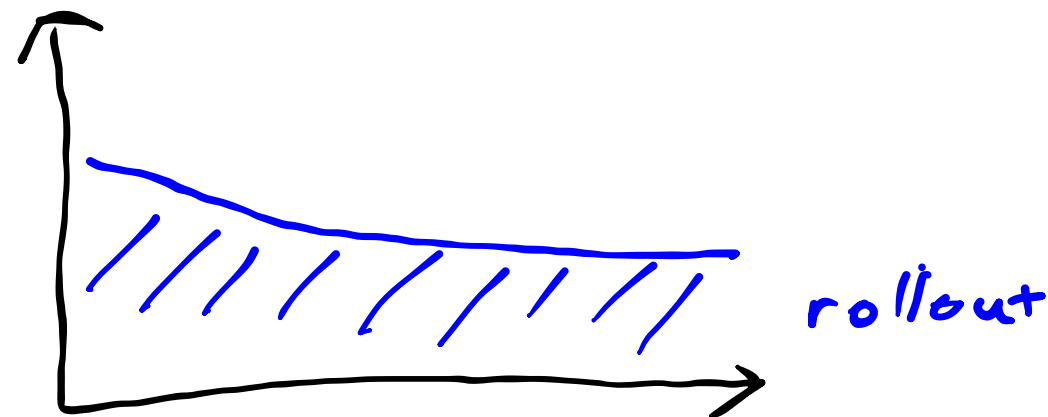
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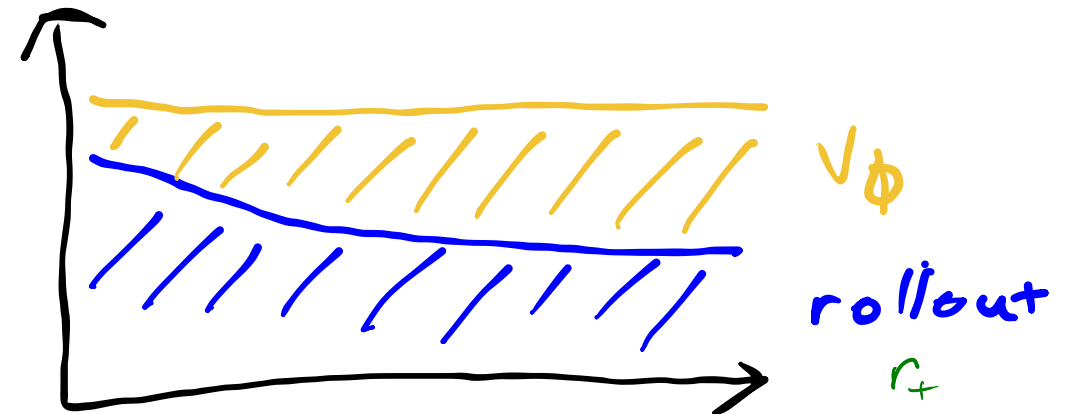
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$\uparrow$  controls rate of rollout trust decay





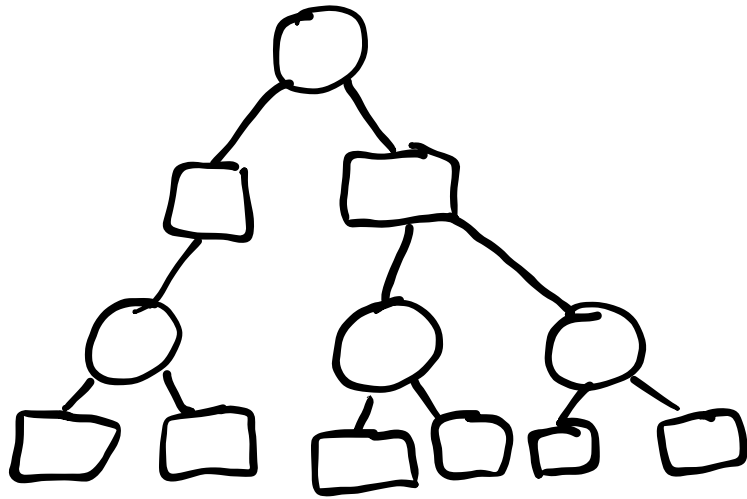
# Alpha Zero: Actor Critic with MCTS

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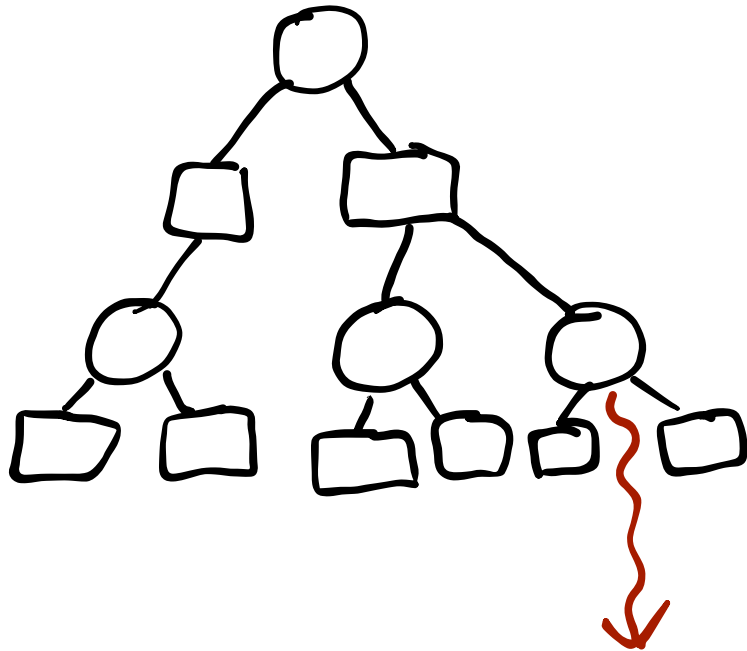
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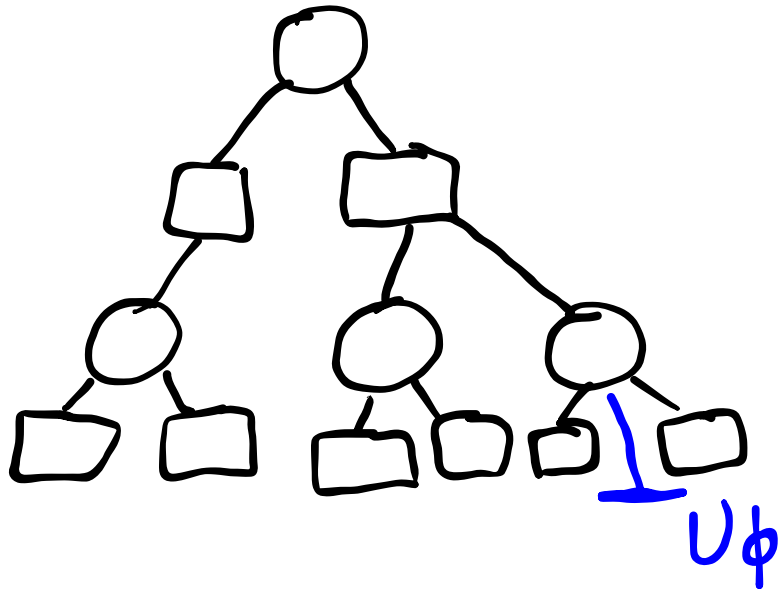
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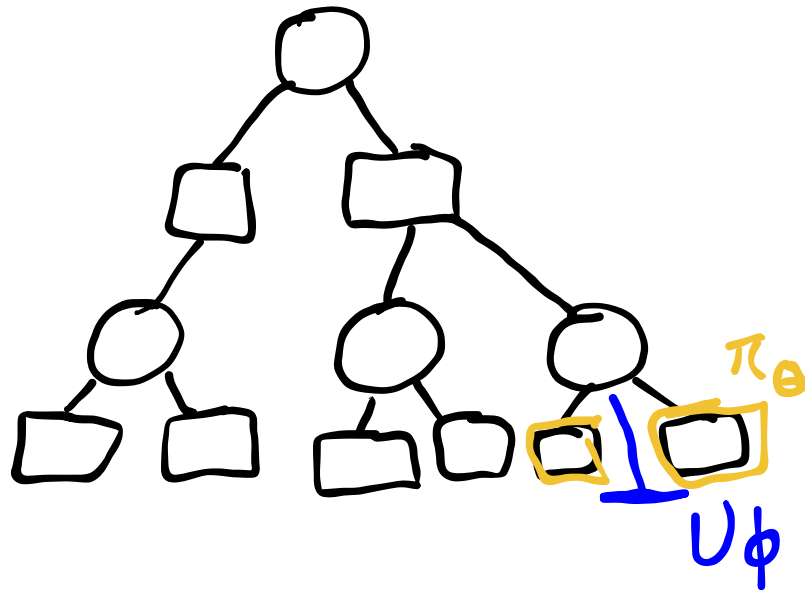
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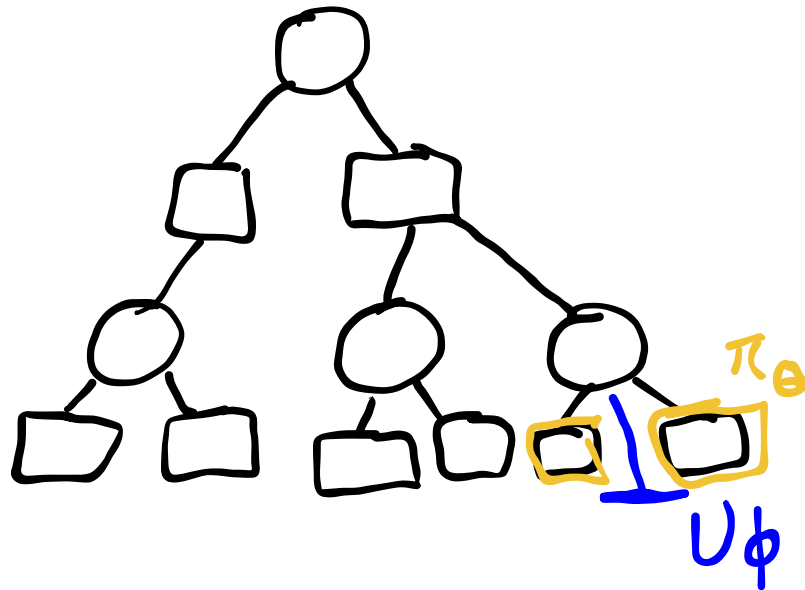
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$$a = \arg \max_a Q(s, a) + c \pi_\theta(a | s) \frac{\sqrt{N(s)}}{1 + N(s, a)}$$

# Alpha Zero: Actor Critic with MCTS

1. Use  $\pi_\theta$  and  $U_\phi$  in MCTS
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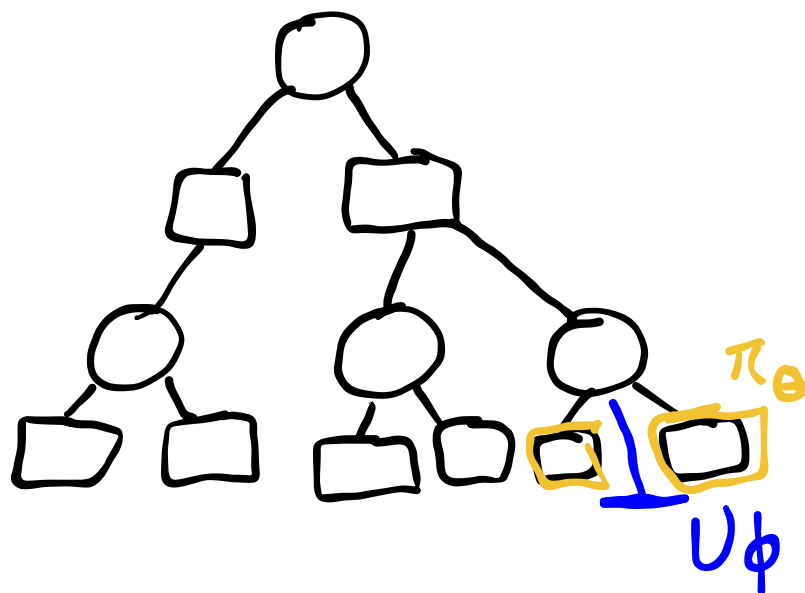
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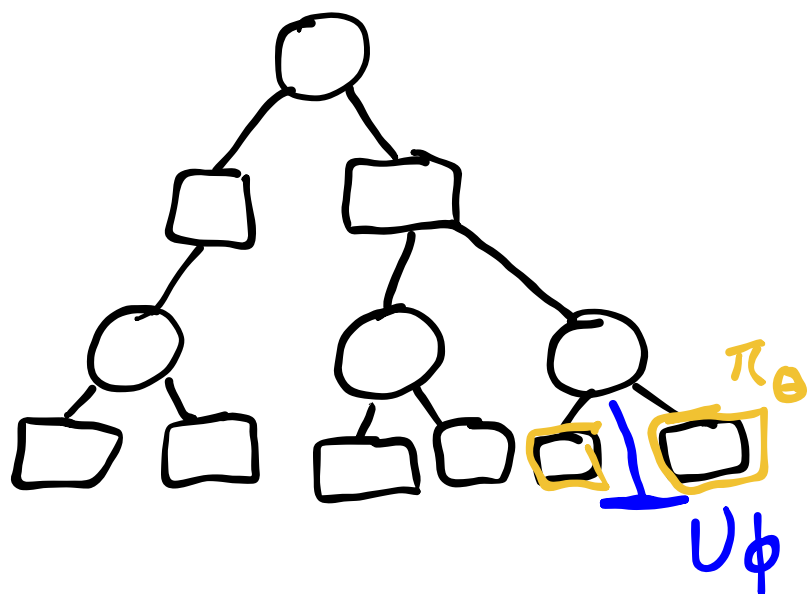


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$$\ell(\phi) = \frac{1}{2} \mathbb{E}_s \left[ (U_\phi(s) - U_{\text{MCTS}}(s))^2 \right]$$

$$U_{\text{MCTS}}(s) = \max_a Q(s, a)$$

$$a = \arg \max_a Q(s, a) + c \pi_\theta(a | s) \frac{\sqrt{N(s)}}{1 + N(s, a)}$$

## 2. Reward Shaping

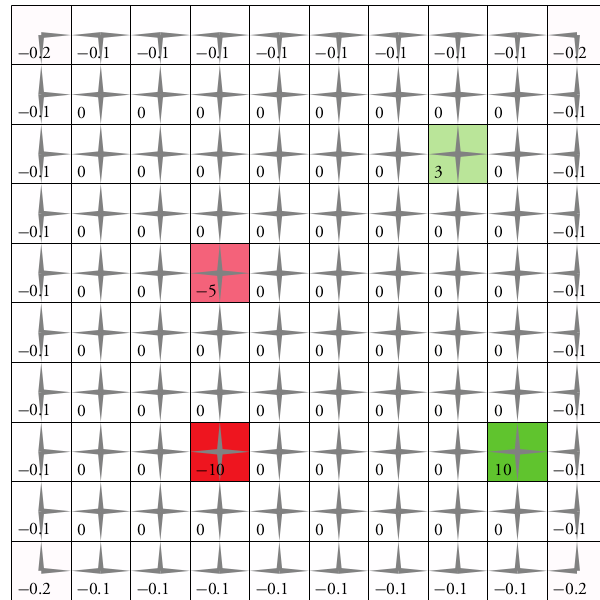
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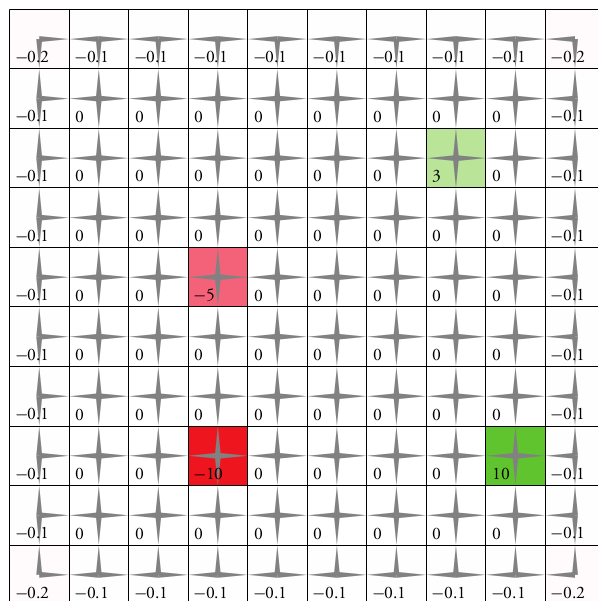
Sparse Reward



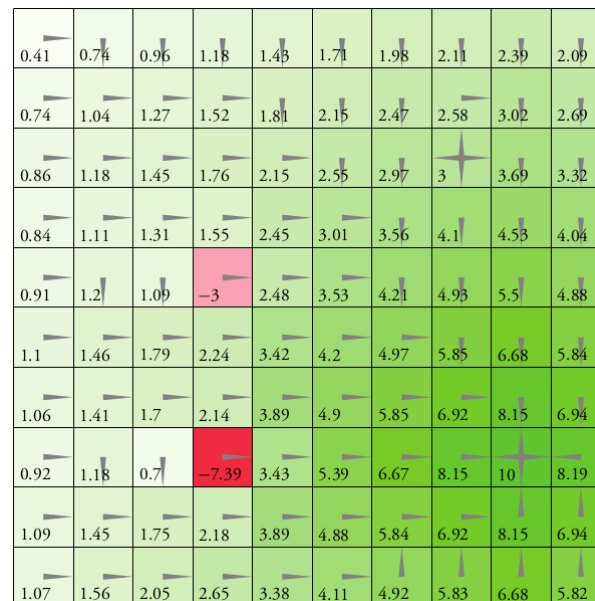
# 2. Reward Shaping

Which is easier to learn on?

## Sparse Reward



## Dense Reward



# Coast Runners 7: A Cautionary Tale



CoastRunners 7

<https://www.youtube.com/watch?v=tlOlHko8ySg>

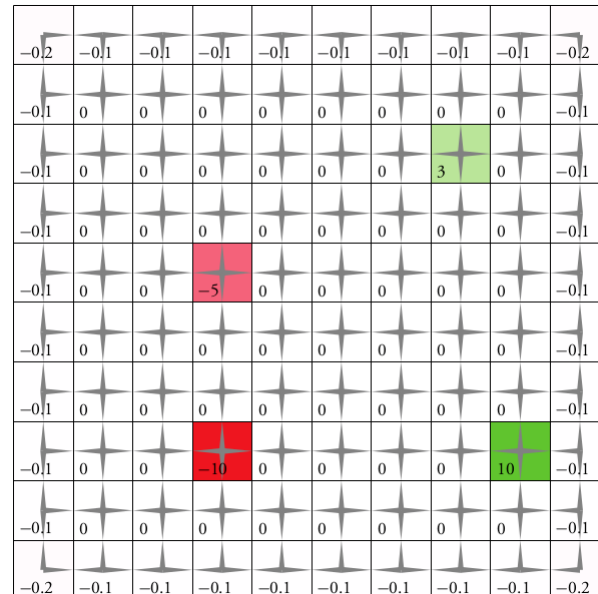
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Reward





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Reward

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
|      |      |      |      |      |      |      |      |      |      |
| -0.2 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 | -0.2 |
|      |      |      |      |      |      |      |      |      |      |
| -0.1 | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | -0.1 |
|      |      |      |      |      |      |      |      |      |      |
| -0.1 | 0    | 0    | 0    | 0    | 0    | 0    | 3    | 0    | -0.1 |
|      |      |      |      |      |      |      |      |      |      |
| -0.1 | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | -0.1 |
|      |      |      |      |      |      |      |      |      |      |
| -0.1 | 0    | 0    | -5   | 0    | 0    | 0    | 0    | 0    | -0.1 |
|      |      |      |      |      |      |      |      |      |      |
| -0.1 | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | -0.1 |
|      |      |      |      |      |      |      |      |      |      |
| -0.1 | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | -0.1 |
|      |      |      |      |      |      |      |      |      |      |
| -0.1 | 0    | 0    |      | 0    | 0    | 0    | 0    | 10   | -0.1 |
|      |      |      |      |      |      |      |      |      |      |
| -0.1 | 0    | 0    | 0    | 0    | 0    | 0    | 0    | 0    | -0.1 |
|      |      |      |      |      |      |      |      |      |      |
| -0.2 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 | -0.1 | -0.2 |

Value

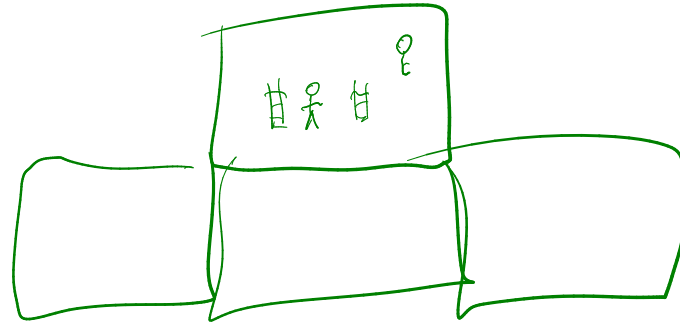
|      |      |      |       |      |      |      |      |      |      |
|------|------|------|-------|------|------|------|------|------|------|
|      |      |      |       |      |      |      |      |      |      |
| 0.41 | 0.74 | 0.96 | 1.18  | 1.45 | 1.71 | 1.98 | 2.11 | 2.39 | 2.09 |
|      |      |      |       |      |      |      |      |      |      |
| 0.74 | 1.04 | 1.27 | 1.52  | 1.81 | 2.15 | 2.47 | 2.58 | 3.02 | 2.69 |
|      |      |      |       |      |      |      |      |      |      |
| 0.86 | 1.18 | 1.45 | 1.76  | 2.15 | 2.55 | 2.97 | 3    | 3.69 | 3.32 |
|      |      |      |       |      |      |      |      |      |      |
| 0.84 | 1.11 | 1.31 | 1.55  | 2.45 | 3.01 | 3.56 | 4.1  | 4.53 | 4.04 |
|      |      |      |       |      |      |      |      |      |      |
| 0.91 | 1.2  | 1.09 | -3    | 2.48 | 3.53 | 4.21 | 4.93 | 5.5  | 4.88 |
|      |      |      |       |      |      |      |      |      |      |
| 1.1  | 1.46 | 1.79 | 2.24  | 3.42 | 4.2  | 4.97 | 5.85 | 6.68 | 5.84 |
|      |      |      |       |      |      |      |      |      |      |
| 1.06 | 1.41 | 1.7  | 2.14  | 3.89 | 4.9  | 5.85 | 6.92 | 8.15 | 6.94 |
|      |      |      |       |      |      |      |      |      |      |
| 0.92 | 1.18 | 0.7  | -7.39 | 3.43 | 5.39 | 6.67 | 8.15 | 10   | 8.19 |
|      |      |      |       |      |      |      |      |      |      |
| 1.09 | 1.45 | 1.75 | 2.18  | 3.89 | 4.88 | 5.84 | 6.92 | 8.15 | 6.94 |
|      |      |      |       |      |      |      |      |      |      |
| 1.07 | 1.56 | 2.05 | 2.65  | 3.38 | 4.11 | 4.92 | 5.83 | 6.68 | 5.82 |

# Potential-Based Reward Shaping

- $R'(s, a, s') = R(s, a, s') + \gamma\Phi(s') - \Phi(s)$
- any other transformation may yield sub optimal policies unless further assumptions are made about the underlying MDP

# Is Exploration Important?

## Montezuma's Revenge



# Is Exploration Important?

## Theory

|             | Algorithm                                                      | Regret                                            | Time                           | Space                 |
|-------------|----------------------------------------------------------------|---------------------------------------------------|--------------------------------|-----------------------|
| Model-based | UCRL2 [10] <sup>1</sup>                                        | at least $\tilde{\mathcal{O}}(\sqrt{H^4 S^2 AT})$ | $\Omega(T S^2 A)$              | $\mathcal{O}(S^2 AH)$ |
|             | Agrawal and Jia [1] <sup>1</sup>                               | at least $\tilde{\mathcal{O}}(\sqrt{H^3 S^2 AT})$ |                                |                       |
|             | UCBVI [5] <sup>2</sup>                                         | $\tilde{\mathcal{O}}(\sqrt{H^2 SAT})$             | $\tilde{\mathcal{O}}(T S^2 A)$ |                       |
|             | vUCQ [12] <sup>2</sup>                                         | $\tilde{\mathcal{O}}(\sqrt{H^2 SAT})$             |                                |                       |
| Model-free  | Q-learning ( $\varepsilon$ -greedy) [14]<br>(if 0 initialized) | $\Omega(\min\{T, A^{H/2}\})$                      | $\mathcal{O}(T)$               | $\mathcal{O}(SAH)$    |
|             | Delayed Q-learning [25] <sup>3</sup>                           | $\tilde{\mathcal{O}}_{S,A,H}(T^{4/5})$            |                                |                       |
|             | Q-learning (UCB-H)                                             | $\tilde{\mathcal{O}}(\sqrt{H^4 SAT})$             |                                |                       |
|             | Q-learning (UCB-B)                                             | $\tilde{\mathcal{O}}(\sqrt{H^3 SAT})$             |                                |                       |
|             | lower bound                                                    | $\Omega(\sqrt{H^2 SAT})$                          | -                              | -                     |

Table 1: Regret comparisons for RL algorithms on episodic MDP.  $T = KH$  is totally number of steps,  $H$  is the number of steps per episode,  $S$  is the number of states, and  $A$  is the number of actions. For clarity, this table is presented for  $T \geq \text{poly}(S, A, H)$ , omitting low order terms.

# Exploration Bonus

# Exploration Bonus

- In General,  $R^+(s, a) = R(s, a) + B(s, a)$
- UCB:  $B(s, a) = c\sqrt{\frac{\log N(s)}{N(s, a)}}$

# Exploration Bonus

## Example 1: Learn Pseudocount

Bellemare, et al. 2016 "Unifying Count-Based Exploration..."

# Exploration Bonus

## Example 1: Learn Pseudocount

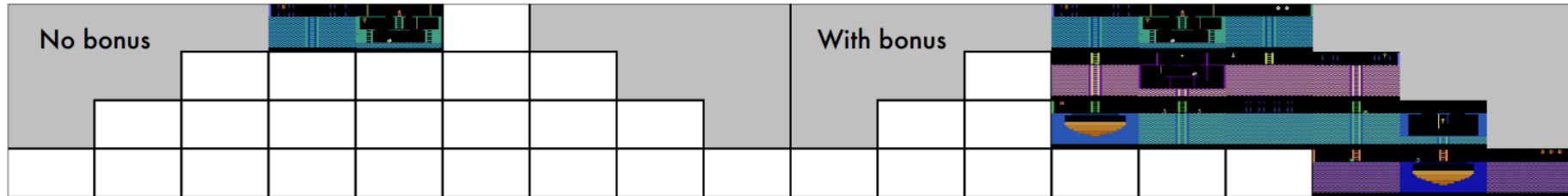
$B(s, a) \approx \frac{1}{\sqrt{\hat{N}(s)}}$  where  $\hat{N}(s)$  is a learned function approximation



# Exploration Bonus

## Example 1: Learn Pseudocount

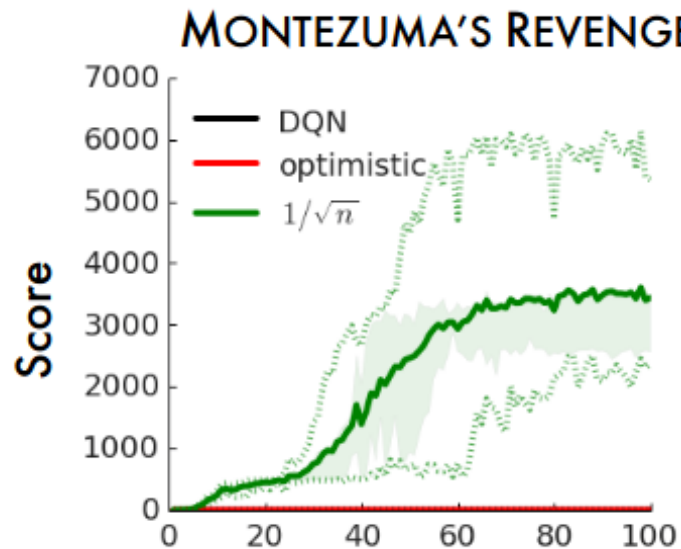
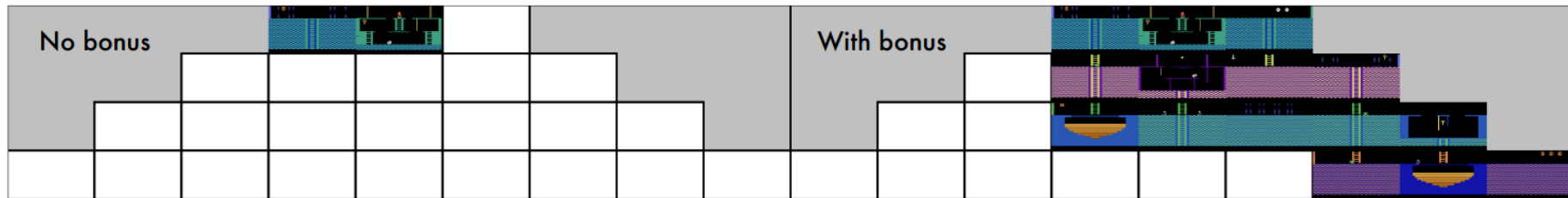
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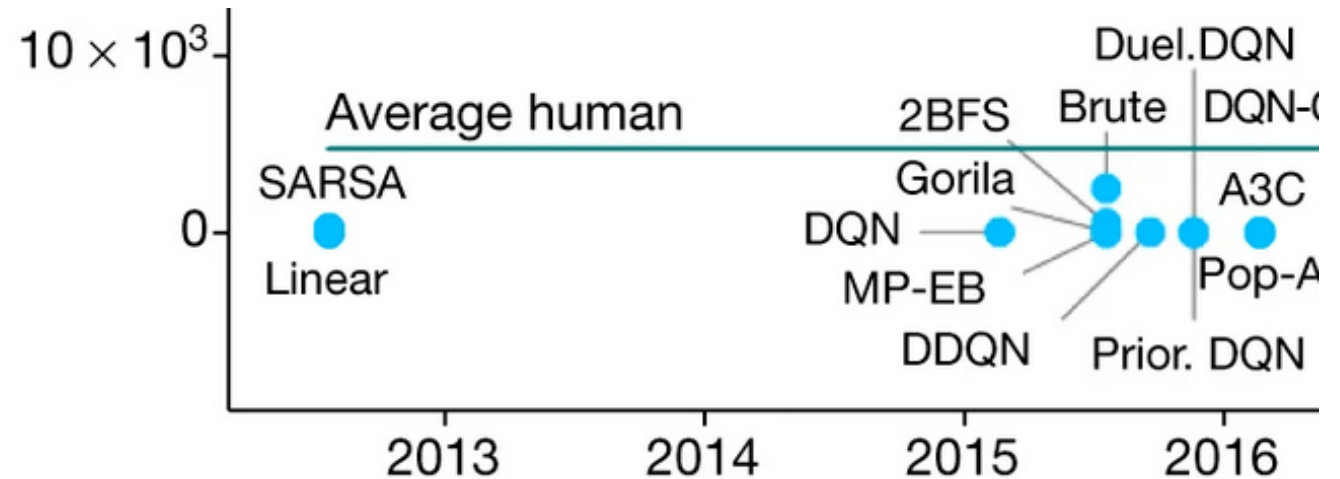
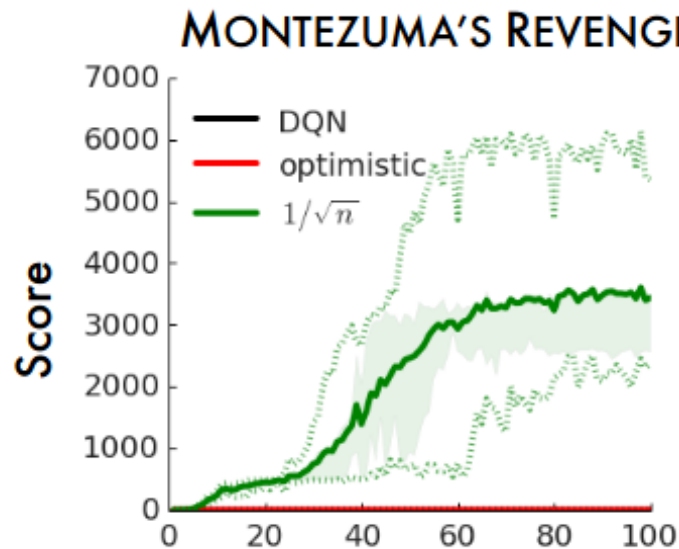
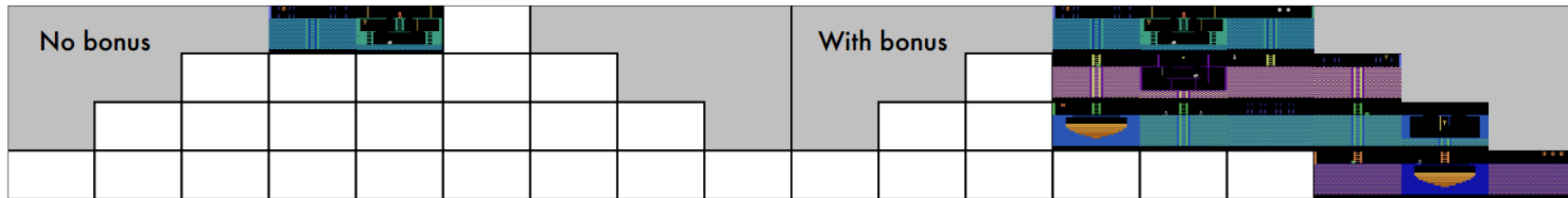


Bellemare, et al. 2016 "Unifying Count-Based Exploration..."

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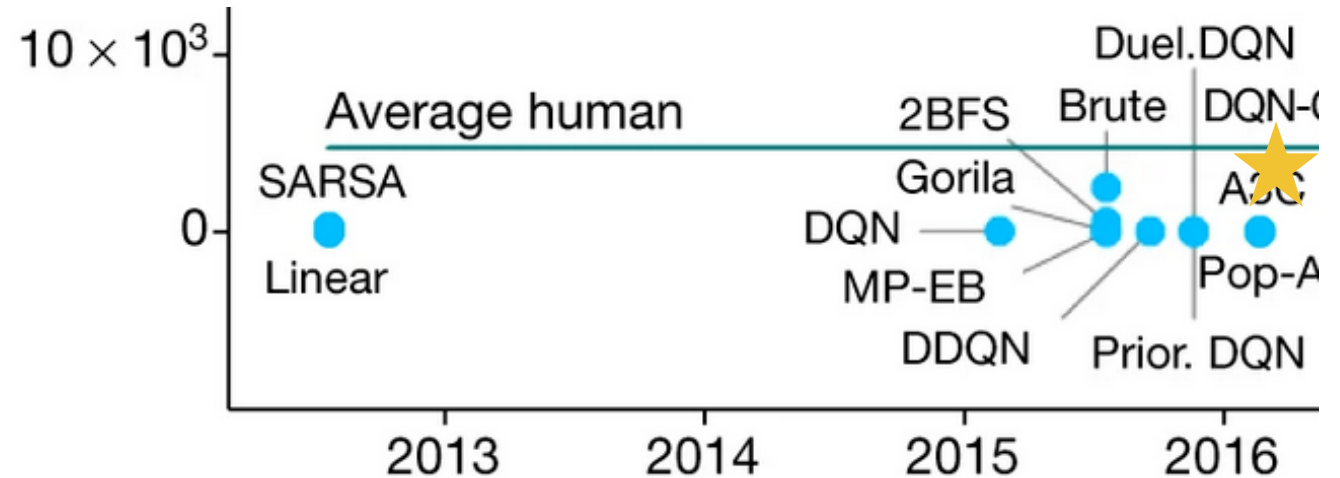
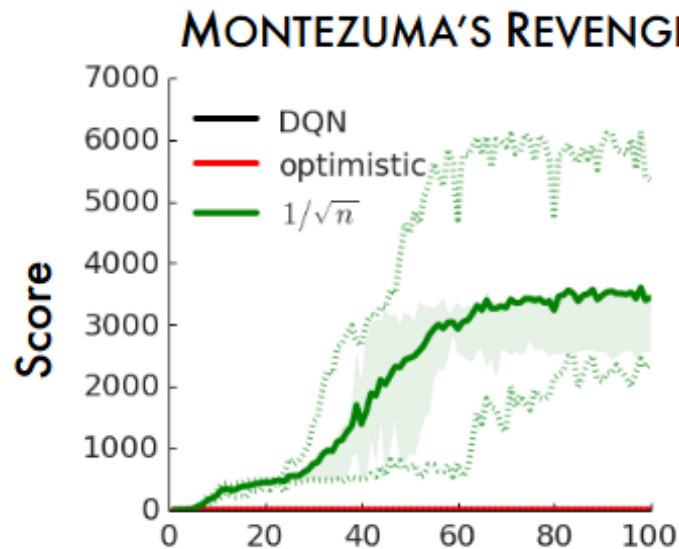
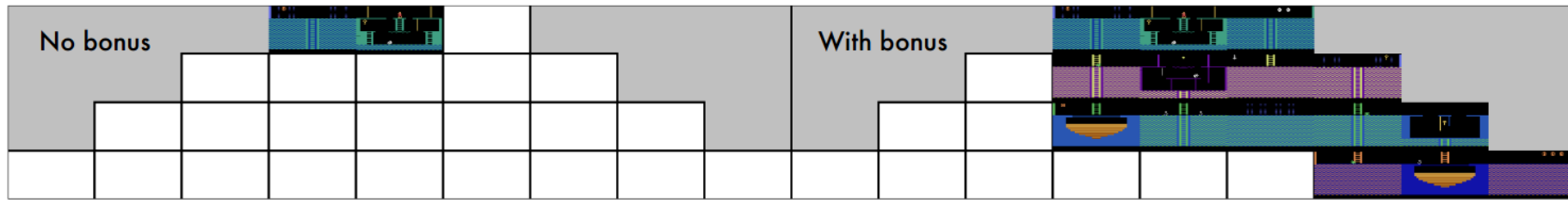


Bellemare, et al. 2016 "Unifying Count-Based Exploration..."

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Bellemare, et al. 2016 "Unifying Count-Based Exploration..."

# Exploration Bonus

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# Exploration Bonus

Example 2: Learn a function of the state and action

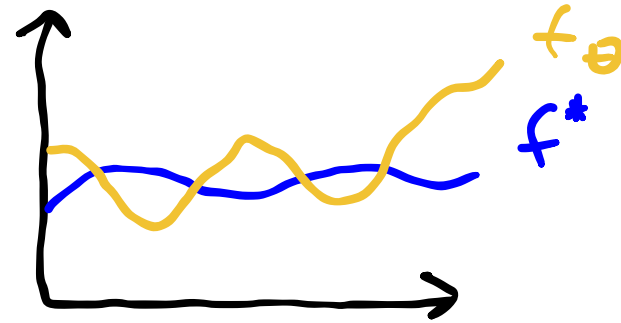
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# Exploration Bonus

Example 2: Learn a function of the state and action

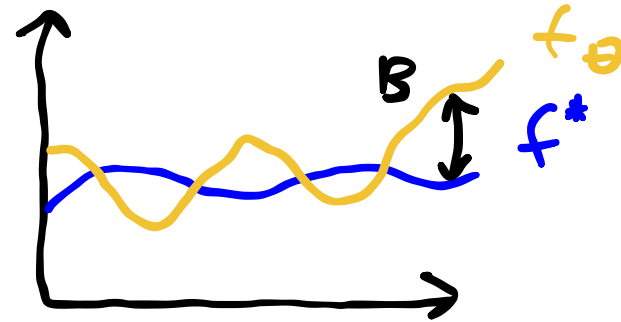
$$B(s, a) = \|\hat{f}_\theta(s, a) - f^*(s, a)\|^2$$



# Exploration Bonus

Example 2: Learn a function of the state and action

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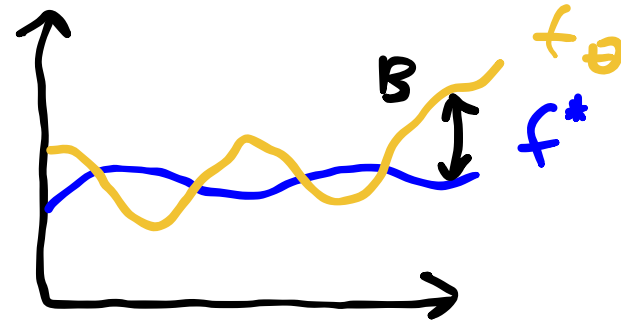


# Exploration Bonus

**Example 2: Learn a function of the state and action**

$$B(s, a) = \|\hat{f}_\theta(s, a) - f^*(s, a)\|^2$$

What should  $f^*$  be?

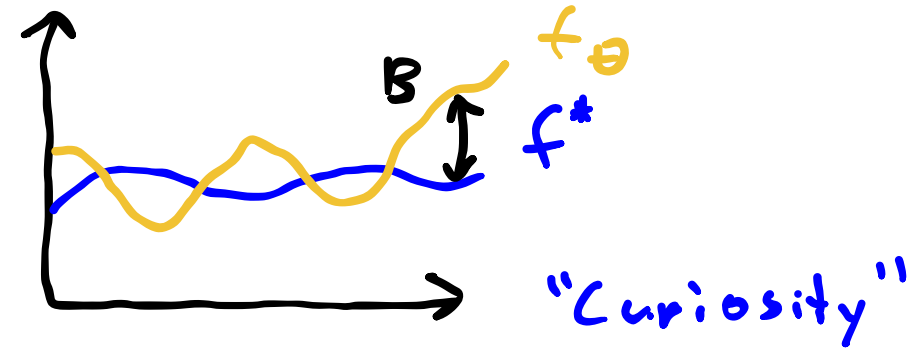


# Exploration Bonus

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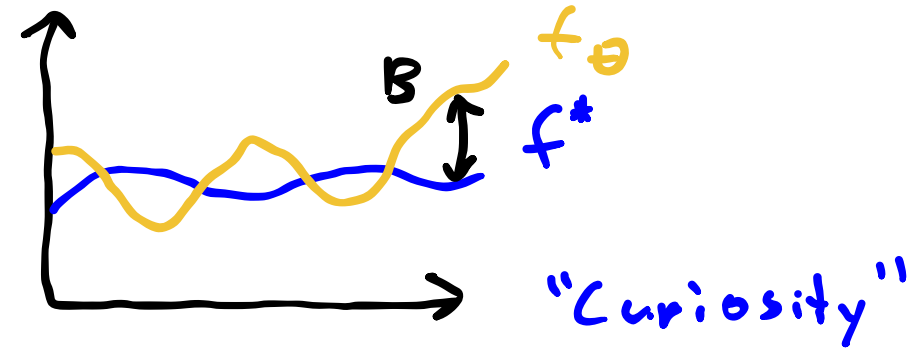
# Exploration Bonus

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What should  $f^*$  be?

- $f^*(s, a) = s'$  (Next state prediction)



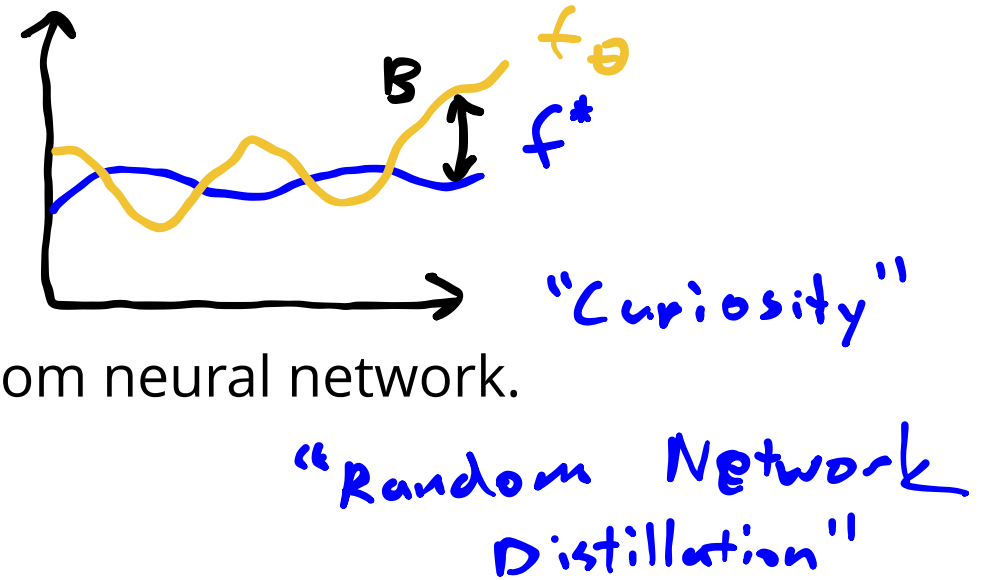
# Exploration Bonus

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$$B(s, a) = \|\hat{f}_\theta(s, a) - f^*(s, a)\|^2$$

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- $f^*(s, a) = f_\phi(s, a)$  where  $f_\phi$  is a random neural network.



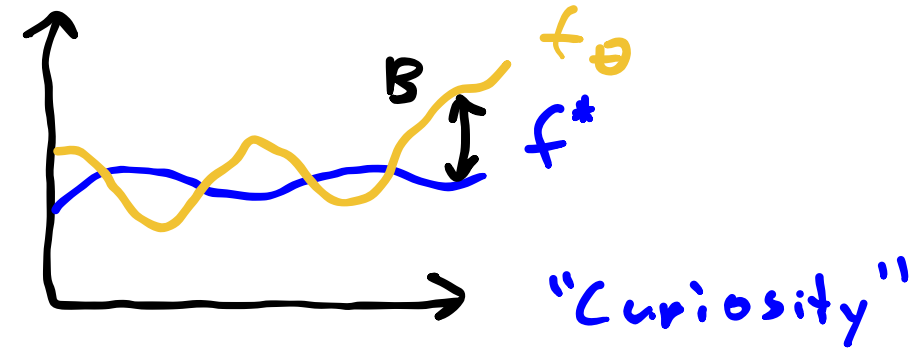
# Exploration Bonus

## Example 2: Learn a function of the state and action

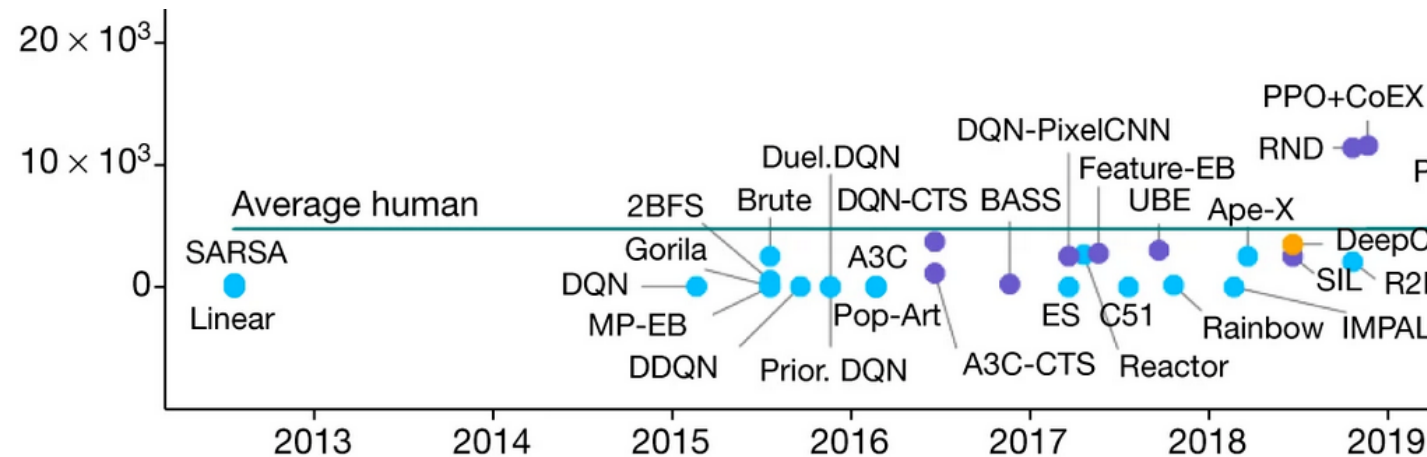
$$B(s, a) = \|\hat{f}_\theta(s, a) - f^*(s, a)\|^2$$

What should  $f^*$  be?

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"Random Network Distillation"



# Exploration Bonus

# Exploration Bonus

## Example 3: Thompson Sampling

# Exploration Bonus

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1. Maintain a distribution over  $Q$



# Exploration Bonus

## Example 3: Thompson Sampling

1. Maintain a distribution over  $Q$
2. Sample  $Q$

# Exploration Bonus

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1. Maintain a distribution over  $Q$
2. Sample  $Q$
3. Act according to  $Q$

# Exploration Bonus

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1. Maintain a distribution over  $Q$  ← Hard
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# Exploration Bonus

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- Bootstrapping with multiple  $Q$  networks

# Exploration Bonus

## Example 3: Thompson Sampling

1. Maintain a distribution over  $Q$  ← Hard
2. Sample  $Q$
3. Act according to  $Q$

- Bootstrapping with multiple  $Q$  networks
- Dropout

# Exploration Bonus

# Exploration Bonus

## Example 4: Go-Explore

# Exploration Bonus

## Example 4: Go-Explore

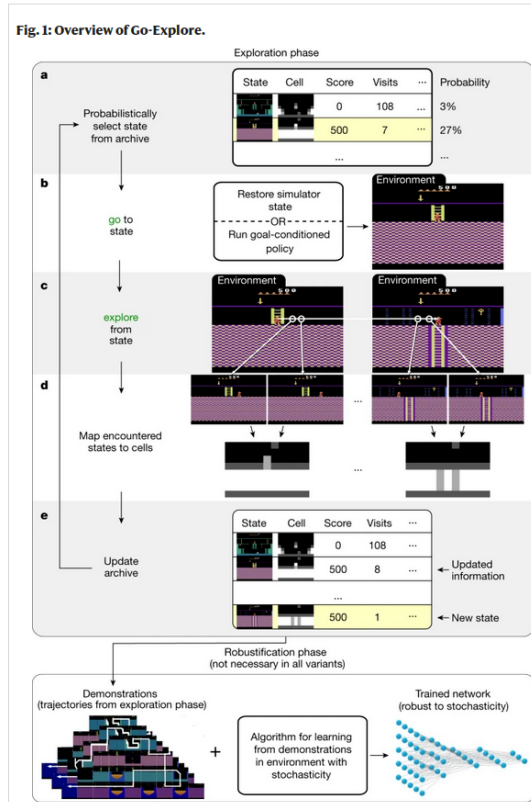
"First return, then explore"



# Exploration Bonus

## Example 4: Go-Explore

"First return, then explore"

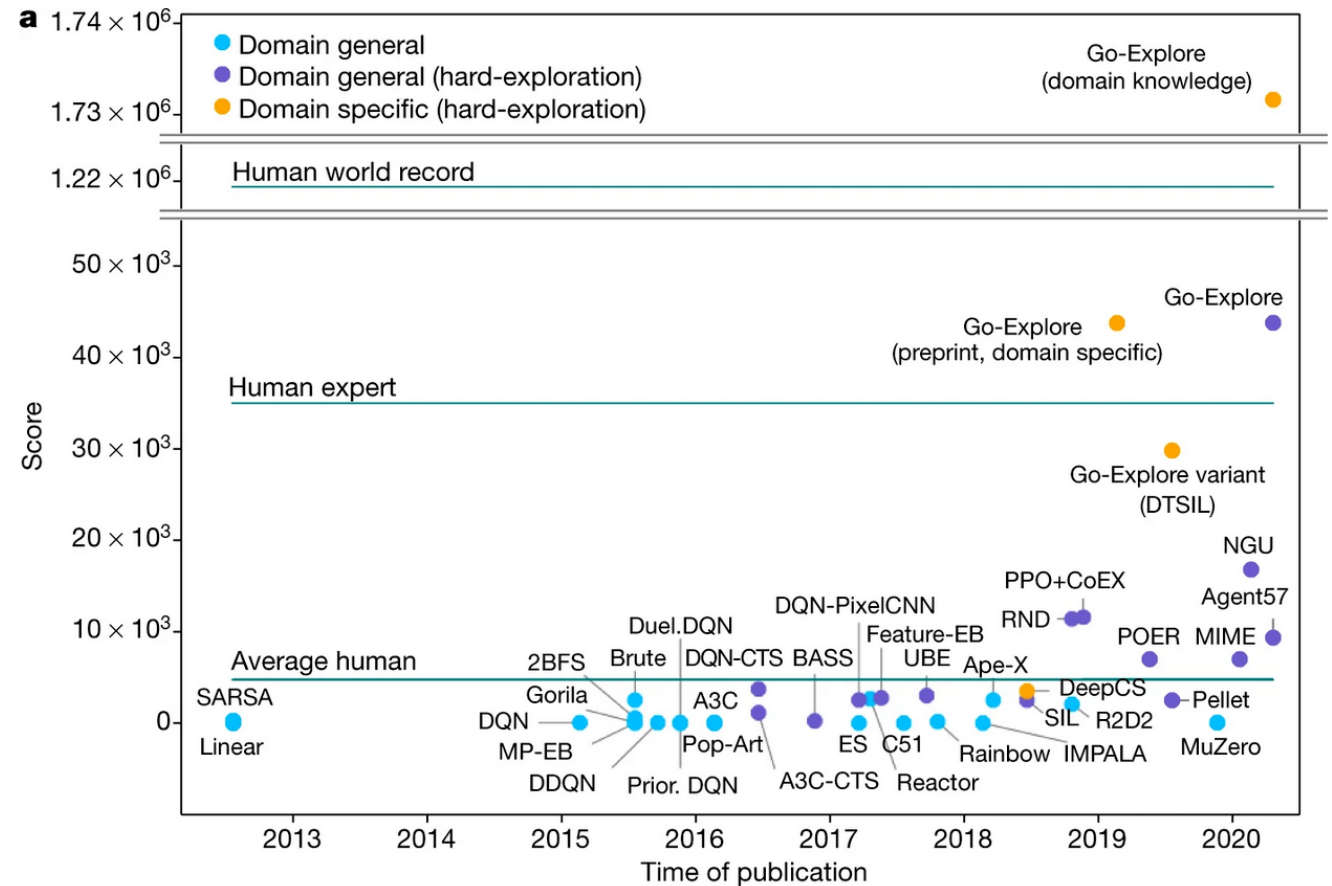
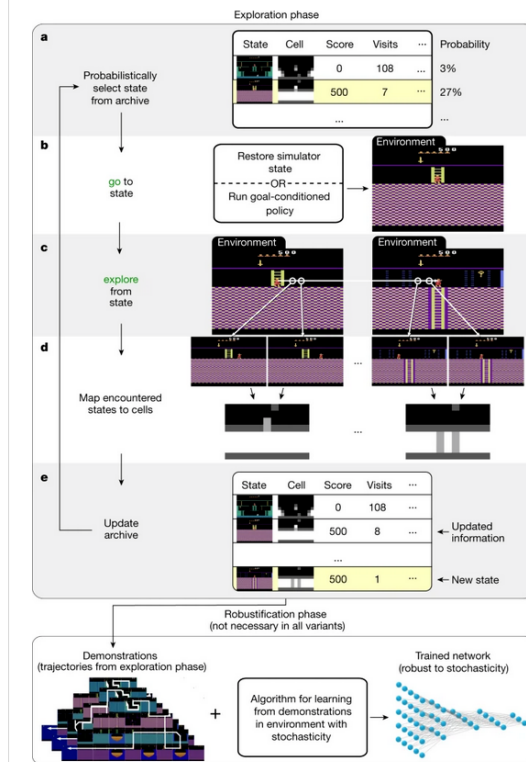


# Exploration Bonus

## Example 4: Go-Explore

"First return, then explore"

Fig. 1: Overview of Go-Explore.

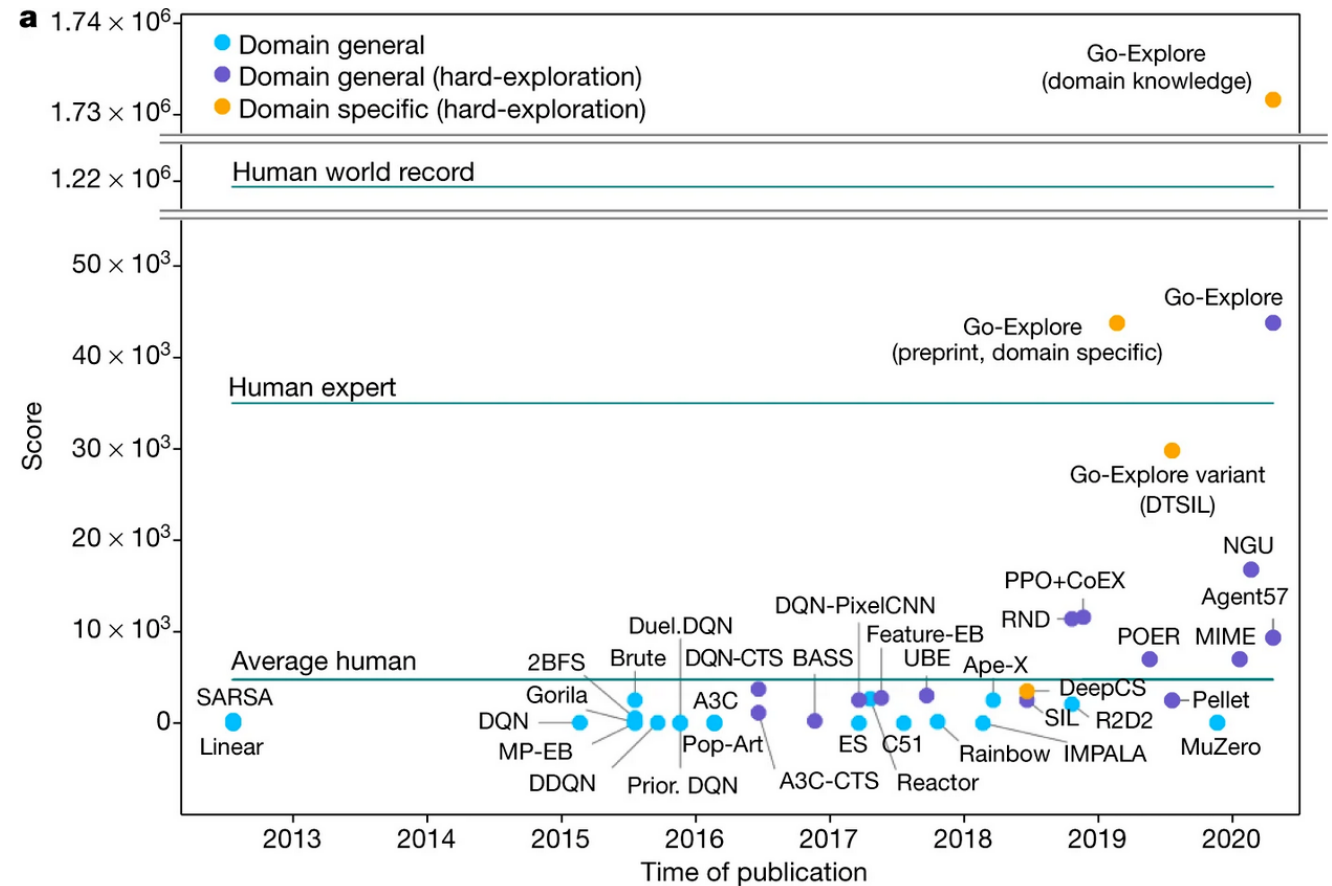
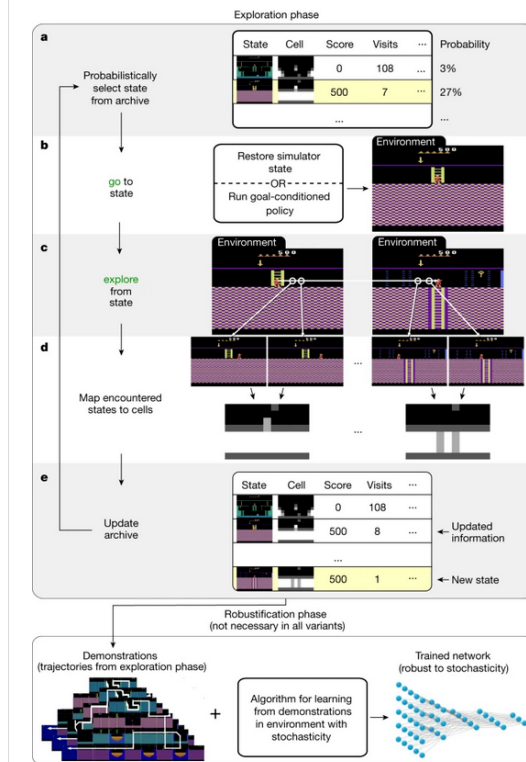


# Exploration Bonus

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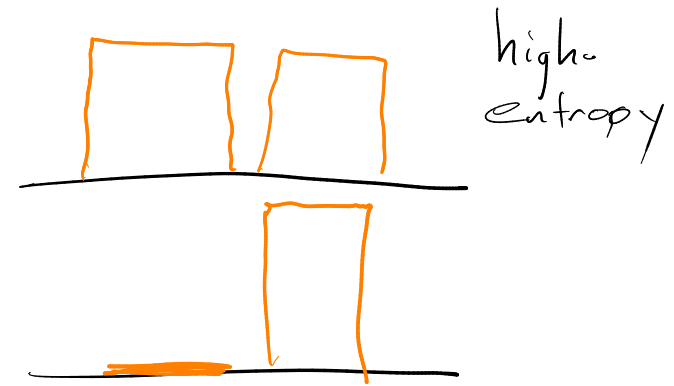
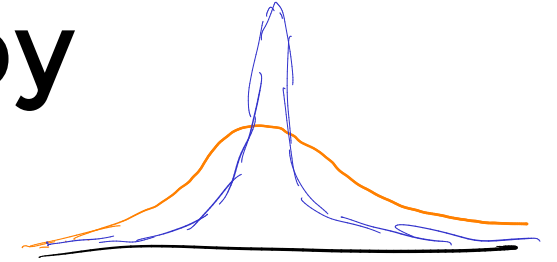


(Uber AI Labs)

# Soft Actor Critic: Entropy Regularization

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$$U(\pi) = E \left[ \sum_{t=0}^{\infty} \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot | s_t))) \right]$$



# Soft Actor Critic: Entropy Regularization

$$U(\pi) = E \left[ \sum_{t=0}^{\infty} \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot | s_t))) \right]$$

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# Soft Actor Critic: Entropy Regularization

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$$\mathcal{T}^{\pi} Q(\mathbf{s}_t, \mathbf{a}_t) \triangleq r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \mathbb{E}_{\mathbf{s}_{t+1} \sim p} [V(\mathbf{s}_{t+1})]$$

# Soft Actor Critic: Entropy Regularization

maximize

$$U(\pi) = E \left[ \sum_{t=0}^{\infty} \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot | s_t))) \right]$$

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$$\pi_{\text{new}} = \arg \min_{\pi' \in \Pi} D_{\text{KL}} \left( \pi'(\cdot | \mathbf{s}_t) \parallel \frac{\exp(Q^{\pi_{\text{old}}}(\mathbf{s}_t, \cdot))}{Z^{\pi_{\text{old}}}(\mathbf{s}_t)} \right)$$



# Soft Actor Critic

# Soft Actor Critic

---

**Algorithm 1** Soft Actor-Critic

---

Initialize parameter vectors  $\psi, \bar{\psi}, \theta, \phi$ .

**for** each iteration **do**

**for** each environment step **do**

$$\mathbf{a}_t \sim \pi_\phi(\mathbf{a}_t | \mathbf{s}_t)$$

$$\mathbf{s}_{t+1} \sim p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)$$

$$\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{s}_t, \mathbf{a}_t, r(\mathbf{s}_t, \mathbf{a}_t), \mathbf{s}_{t+1})\}$$

**end for**

**for** each gradient step **do**

$$\psi \leftarrow \psi - \lambda_V \hat{\nabla}_\psi J_V(\psi)$$

$$\theta_i \leftarrow \theta_i - \lambda_Q \hat{\nabla}_{\theta_i} J_Q(\theta_i) \text{ for } i \in \{1, 2\}$$

$$\phi \leftarrow \phi - \lambda_\pi \hat{\nabla}_\phi J_\pi(\phi)$$

$$\bar{\psi} \leftarrow \tau \psi + (1 - \tau) \bar{\psi}$$

**end for**

**end for**

---

# Soft Actor Critic

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$$J_V(\psi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}} \left[ \frac{1}{2} (V_\psi(\mathbf{s}_t) - \mathbb{E}_{\mathbf{a}_t \sim \pi_\phi} [Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \log \pi_\phi(\mathbf{a}_t|\mathbf{s}_t)])^2 \right]$$

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**end for**

**end for**

---

$$J_V(\psi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}} \left[ \frac{1}{2} \left( V_\psi(\mathbf{s}_t) - \mathbb{E}_{\mathbf{a}_t \sim \pi_\phi} [Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \log \pi_\phi(\mathbf{a}_t|\mathbf{s}_t)] \right)^2 \right]$$

$$J_Q(\theta) = \mathbb{E}_{(\mathbf{s}_t, \mathbf{a}_t) \sim \mathcal{D}} \left[ \frac{1}{2} \left( Q_\theta(\mathbf{s}_t, \mathbf{a}_t) - \hat{Q}(\mathbf{s}_t, \mathbf{a}_t) \right)^2 \right]$$

# Soft Actor Critic

---

**Algorithm 1** Soft Actor-Critic

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---

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$$\hat{Q}(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \mathbb{E}_{\mathbf{s}_{t+1} \sim p} [V_{\bar{\psi}}(\mathbf{s}_{t+1})]$$

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$$\hat{Q}(\mathbf{s}_t, \mathbf{a}_t) = r(\mathbf{s}_t, \mathbf{a}_t) + \gamma \mathbb{E}_{\mathbf{s}_{t+1} \sim p} [V_{\bar{\psi}}(\mathbf{s}_{t+1})]$$

$$J_\pi(\phi) = \mathbb{E}_{\mathbf{s}_t \sim \mathcal{D}} \left[ \text{D}_{\text{KL}} \left( \pi_\phi(\cdot | \mathbf{s}_t) \parallel \frac{\exp(Q_\theta(\mathbf{s}_t, \cdot))}{Z_\theta(\mathbf{s}_t)} \right) \right]$$

# Soft Actor Critic

Advantages:

# Soft Actor Critic

Advantages:

- Stable



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- Learns many near-optimal policies

# Soft Actor Critic

Advantages:

- Stable
- Learns many near-optimal policies
- Exploration

# Soft Actor Critic

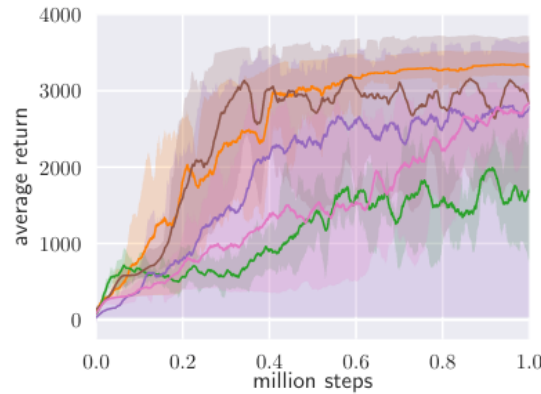
Advantages:

- Stable
- Learns many near-optimal policies
- Exploration
- Insensitivity to hyperparameters
- Off-Policy

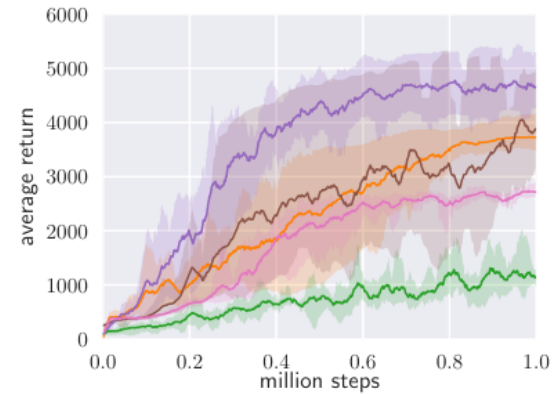
# Soft Actor Critic

Advantages:

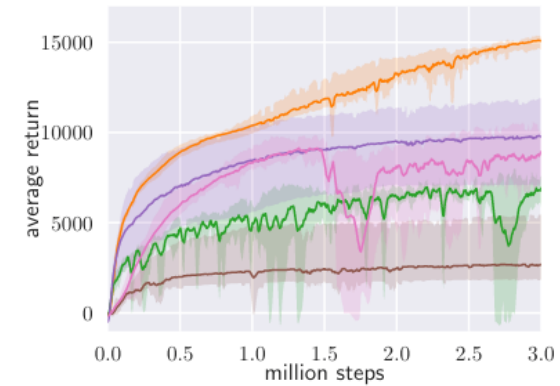
- Stable
- Learns
- Explora
- Insensit
- Off-Poli



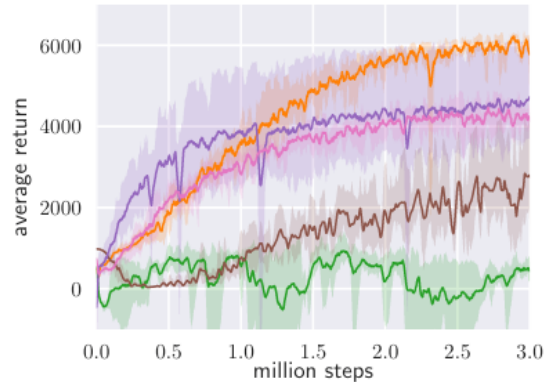
(a) Hopper-v1



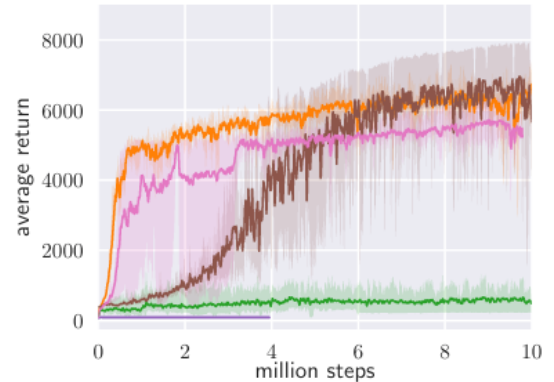
(b) Walker2d-v1



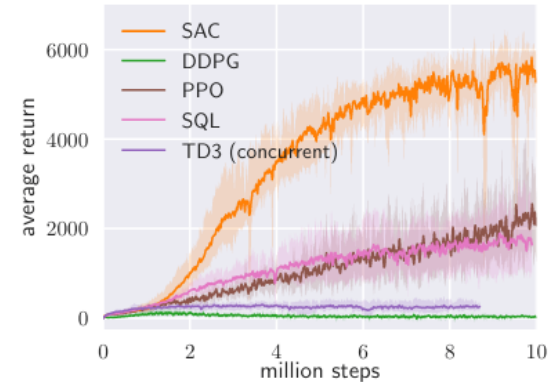
(c) HalfCheetah-v1



(d) Ant-v1



(e) Humanoid-v1



(f) Humanoid (rllab)

# Soft Actor Critic

## Advantages:

- Stable
- Learns many near-optimal policies
- Exploration
- Insensitivity to hyperparameters
- Off-Policy

## Disadvantages

- Sensitive to  $\alpha$  Solution = Entropy  
\*constraint\* and adjust  $\alpha$

# Wisdom

# Deep RL: The Dream

openai / gym

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## Leaderboard

Aman Arora edited this page 3 days ago · 352 revisions

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This page tracks the performance of user algorithms for various tasks in gym. Previously, users could submit their scores directly to [gym.openai.com/envs](https://gym.openai.com/envs), but it has been decided that a simpler wiki might do this task more efficiently.

This wiki page is a community driven page. Anyone can edit this page and add to it. We encourage you to contribute and modify this page and add your scores and links to your write-ups and code to reproduce your results. We also encourage you to add new tasks with the gym interface, but not in the core gym library (such as roboschool) to this page as well.

Links to videos are optional, but encouraged. Videos can be youtube, instagram, a tweet, or other public links. Write-ups should explain how to reproduce the result, and can be in the form of a simple gist link, blog post, or github repo.

We have begun to copy over the previous performance scores and write-up links over from the [previous page](#). This is an ongoing effort, and we can use some help.

## Environments

### Classic control

#### CartPole-v0

A pole is attached by an un-actuated joint to a cart, which moves along a frictionless track. The system is controlled by applying a force of +1 or -1 to the cart. The pendulum starts upright, and the goal is to prevent it from falling over. A reward of +1 is provided for every timestep that the pole remains upright. The episode ends when the pole is more than 15 degrees from vertical, or the cart moves more than 2.4 units from the center.



- [Environment Details](#)
- CartPole-v0 defines "solving" as getting average reward of 195.0 over 100 consecutive trials.*
- This environment corresponds to the version of the cart-pole problem described by Barto, Sutton, and Anderson [Barto83].*

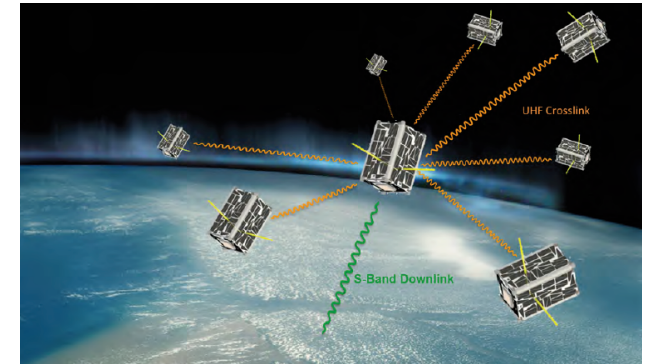
| User          | Episodes before solve            | Write-up                     | Video                 |
|---------------|----------------------------------|------------------------------|-----------------------|
| Zhiqing Xiao  | 0 (use close-form preset policy) | <a href="#">writeup</a>      |                       |
| Hengjian Jia  | 0 (use close-form PID policy)    | <a href="#">code/writeup</a> |                       |
| Keavnn        | 0                                | <a href="#">writeup</a>      |                       |
| Shakti Kumar  | 0                                | <a href="#">writeup</a>      | <a href="#">Video</a> |
| Nextgrid.ai 🍌 | 0                                | <a href="#">writeup</a>      | <a href="#">Video</a> |
| iRyanBell     | 2                                | <a href="#">writeup</a>      |                       |

# Using Deep RL for your problem



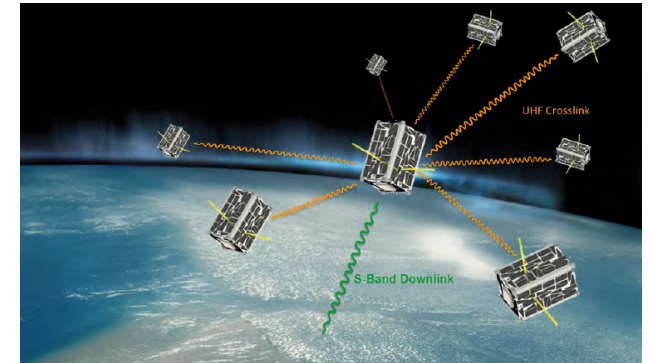
# Using Deep RL for your problem

1. Some interesting problem (smallsat swarm)



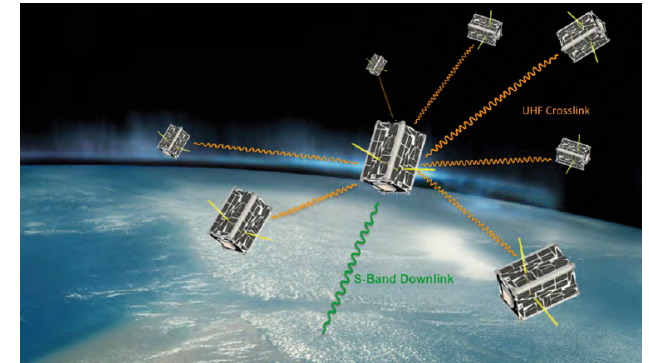
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2. Spend weeks theorizing about the exact-right cost function and dynamics



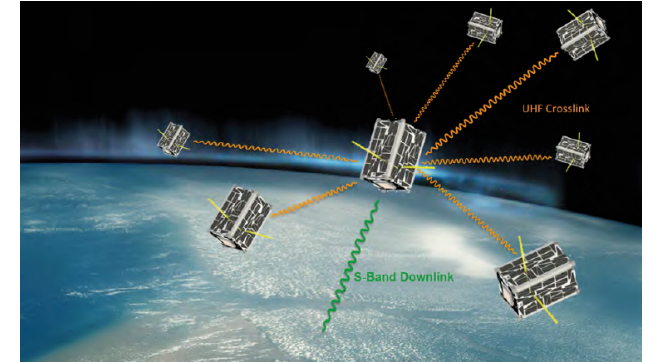
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3. Decide RL can solve all of your problems



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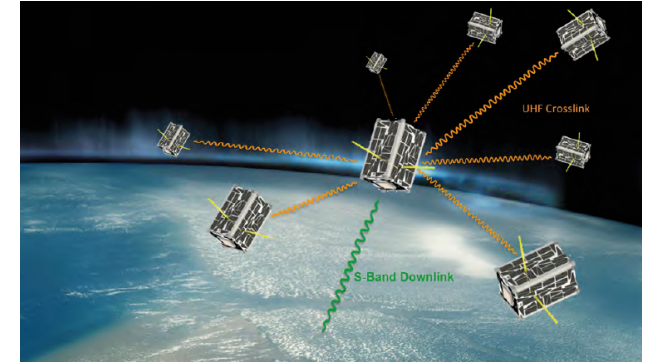
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4. Fire up open-ai baselines



 [openai / baselines](#)

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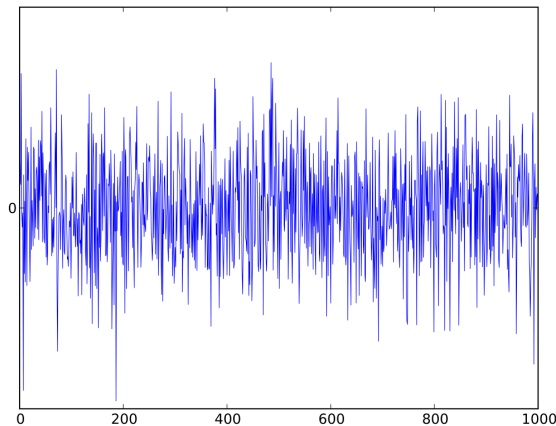
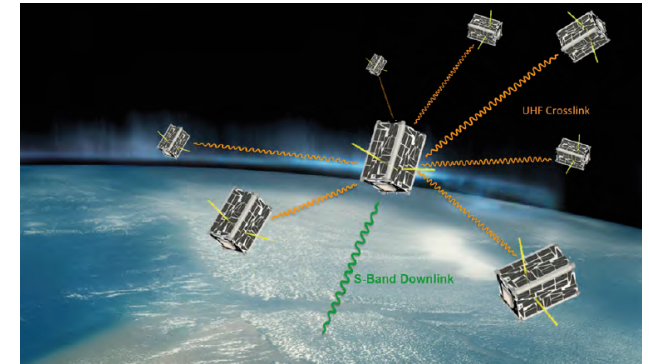
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 [openai / baselines](#)

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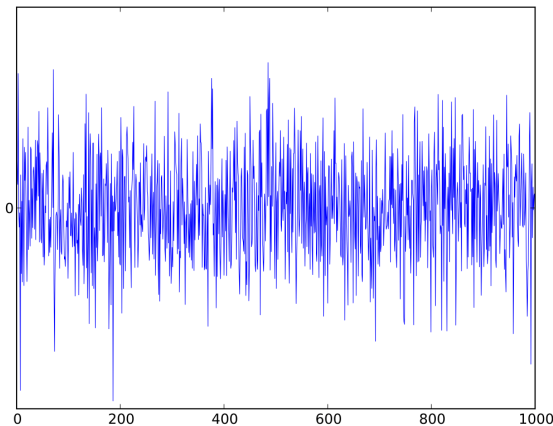
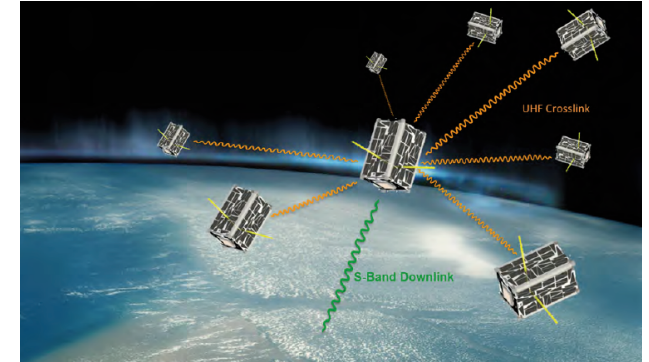
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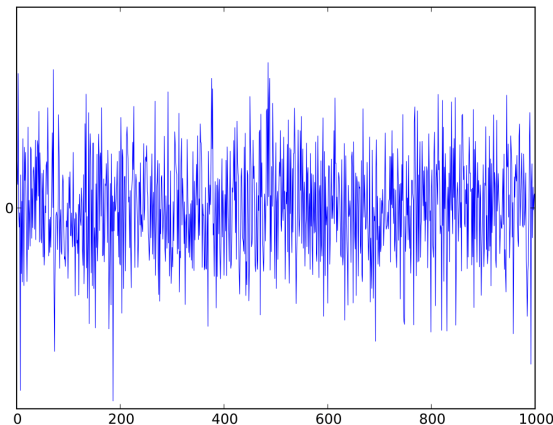
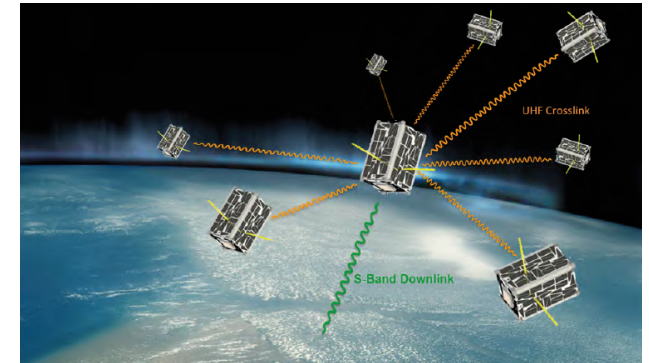


 [openai / baselines](#)

Why not?

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 [openai / baselines](#)

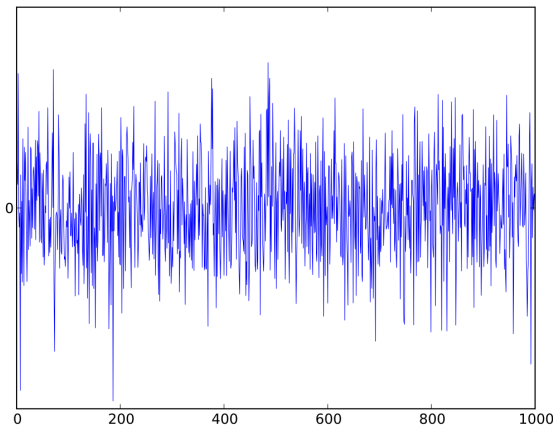
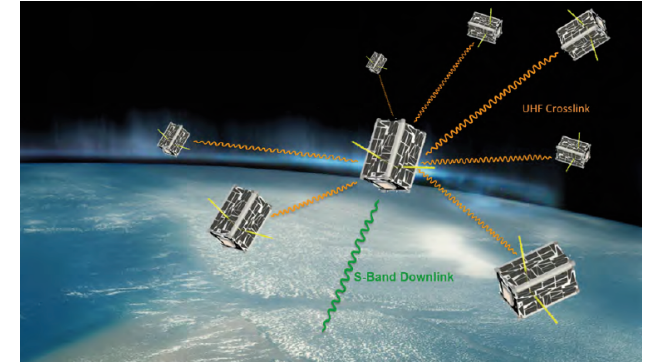
Why not?

- Hyperparameters?



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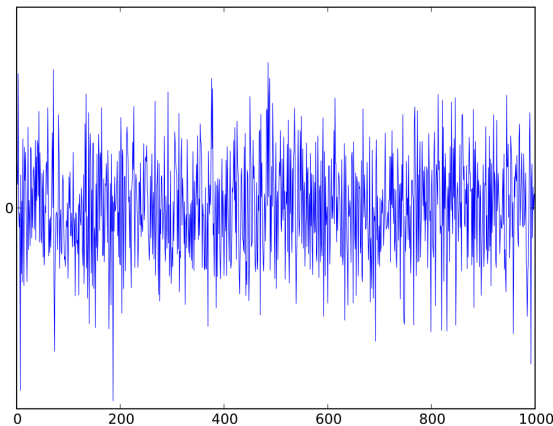
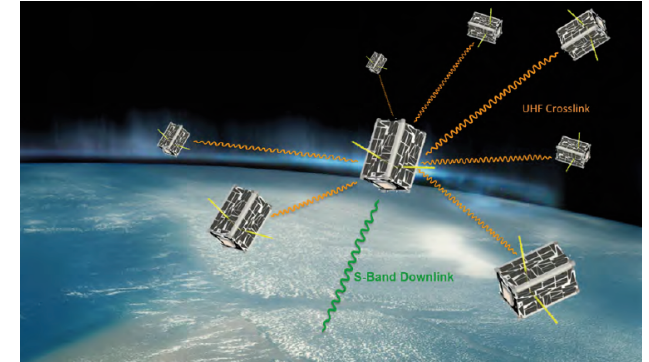
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Why not?

- Hyperparameters?
- Reward scaling?

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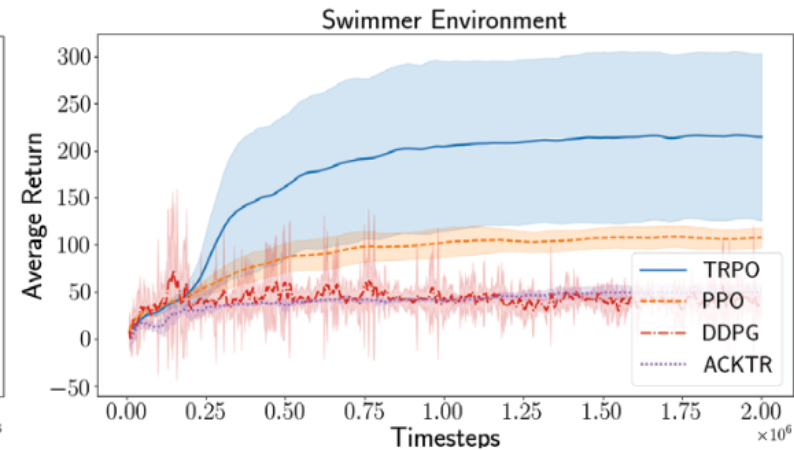
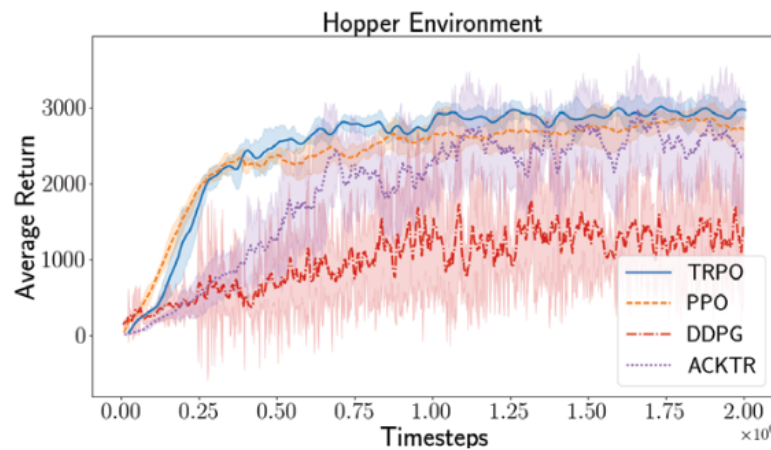
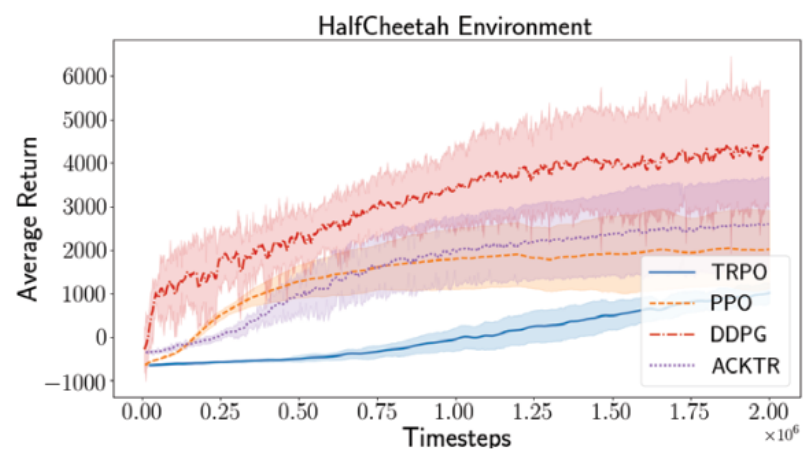


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Why not?

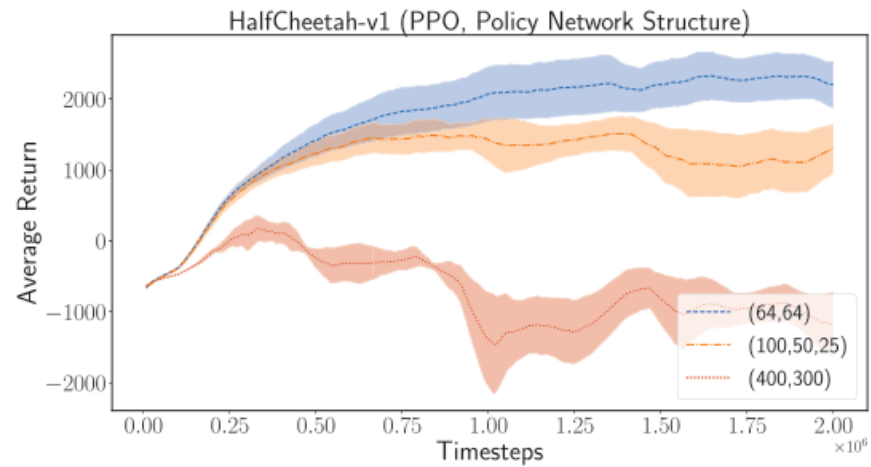
- Hyperparameters?
- Reward scaling?
- Not enough training time????

# Algorithms

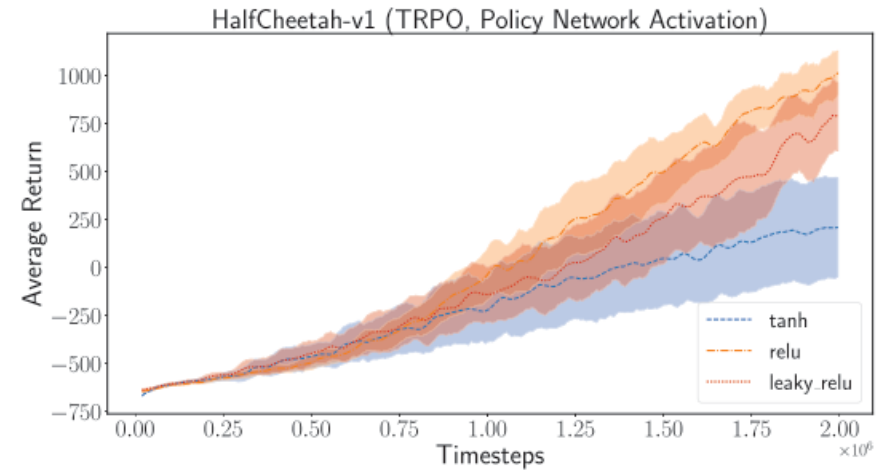
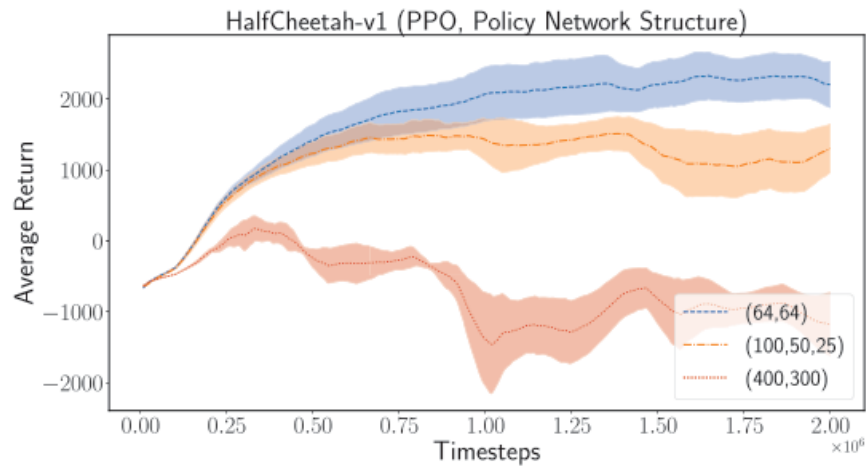


# Policy Network Architecture

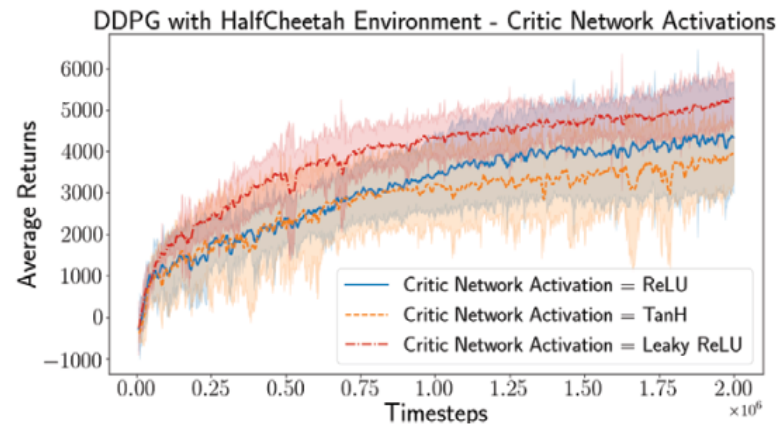
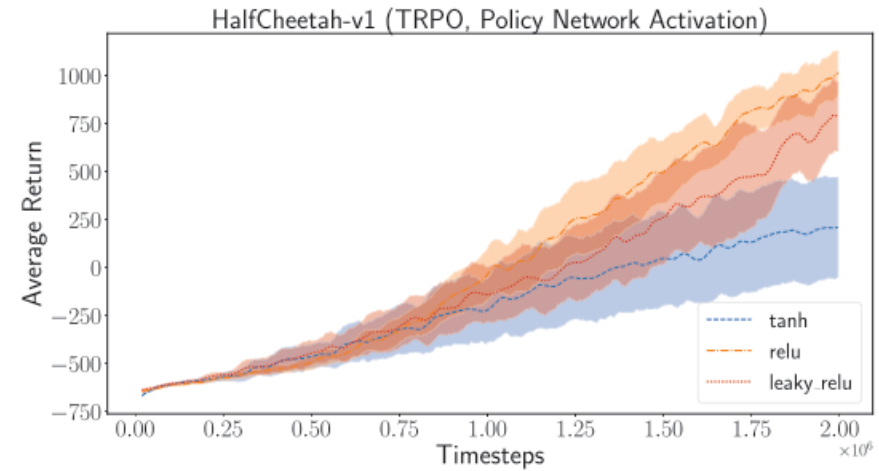
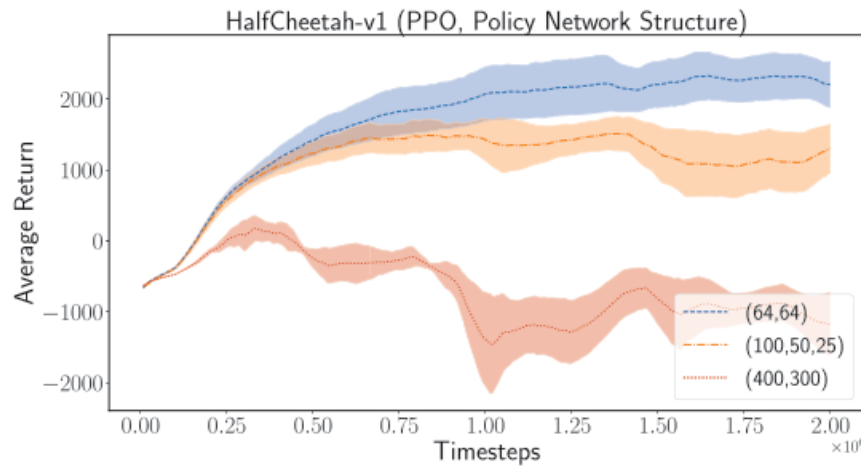
# Policy Network Architecture



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# Reward Rescaling

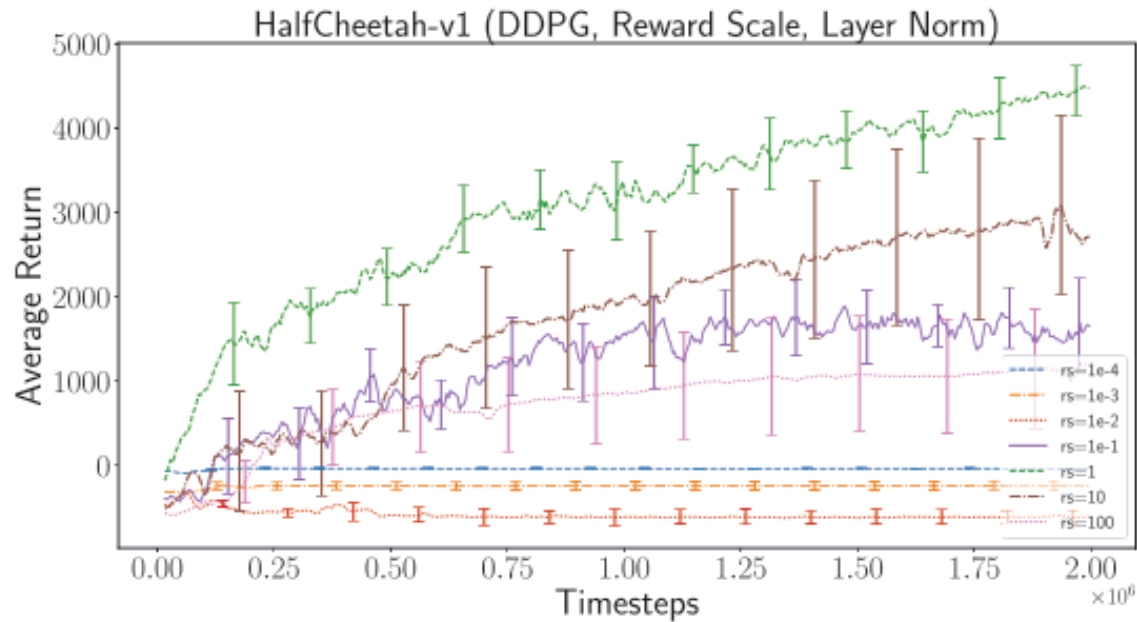


# Reward Rescaling

"simply multiplying the rewards generated from an environment by some scalar"

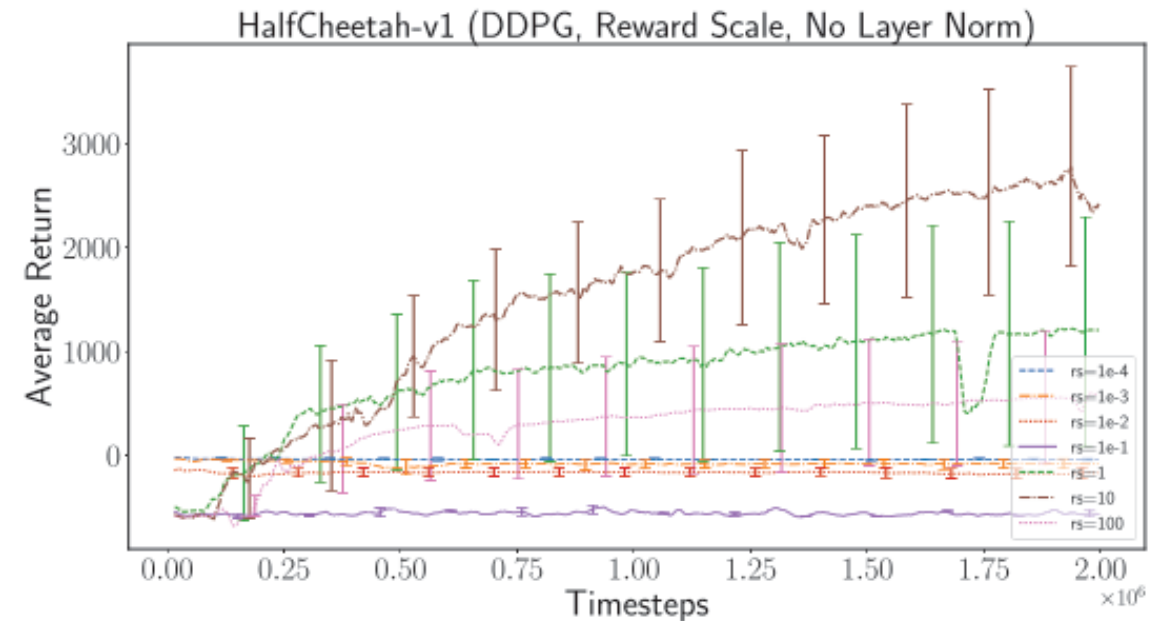
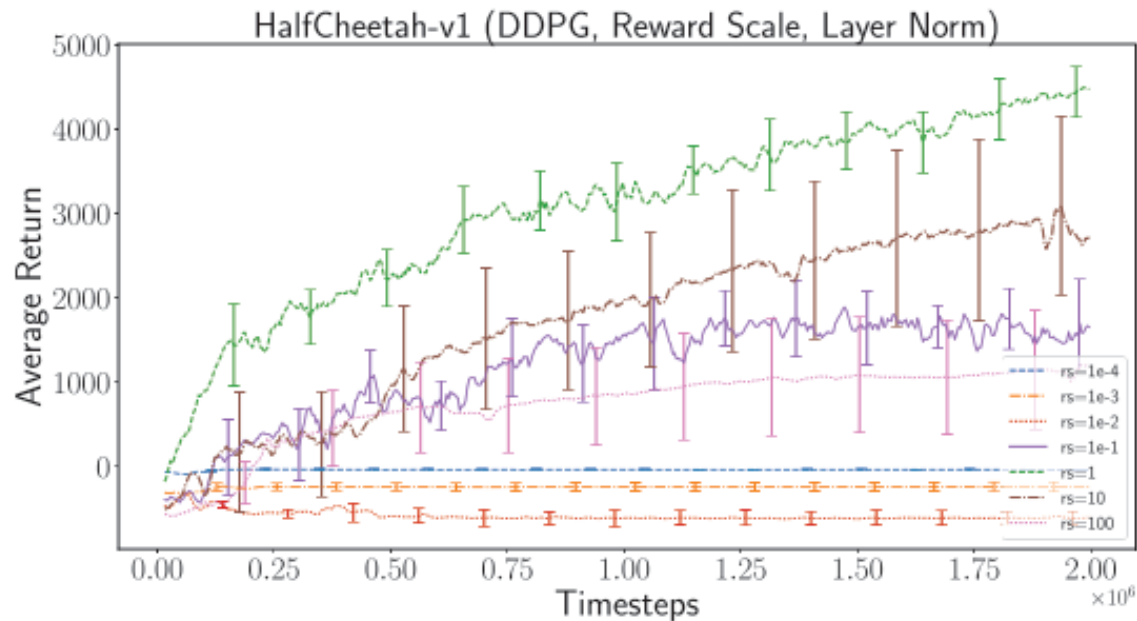
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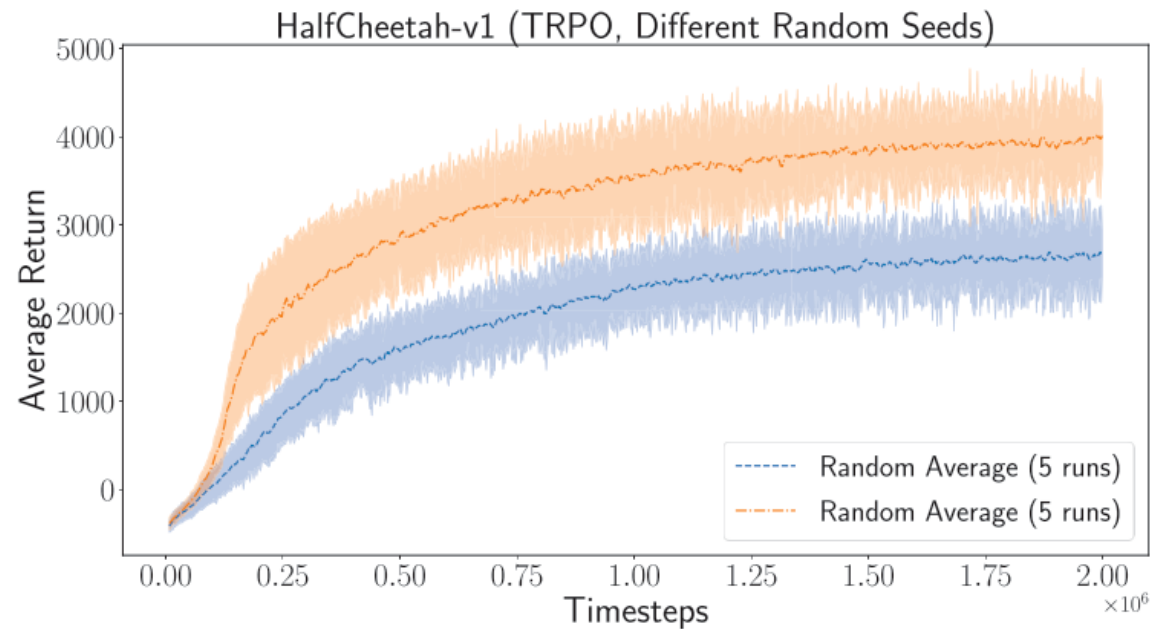
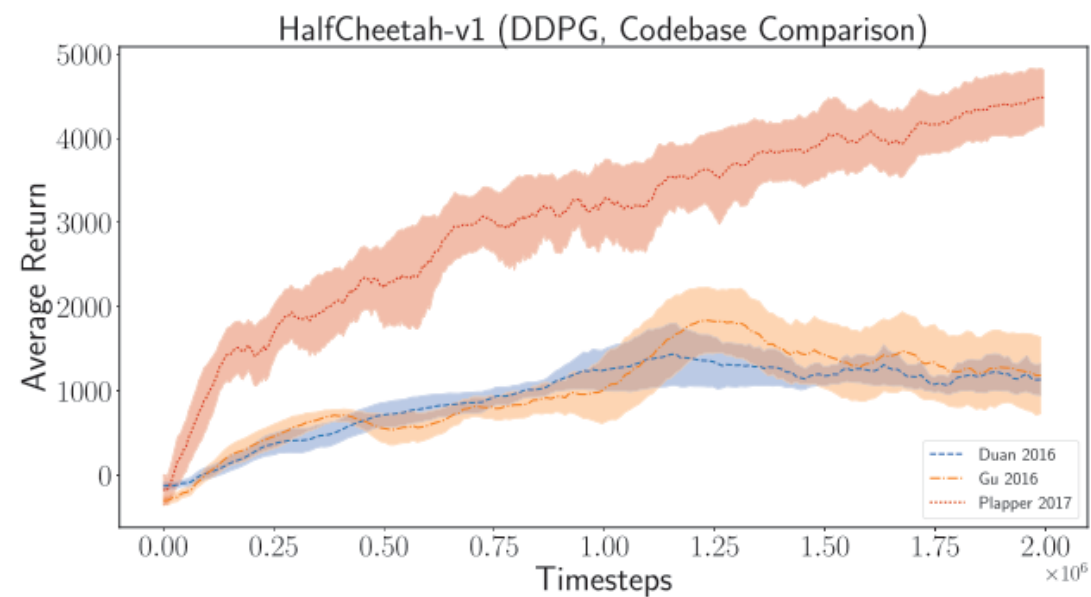
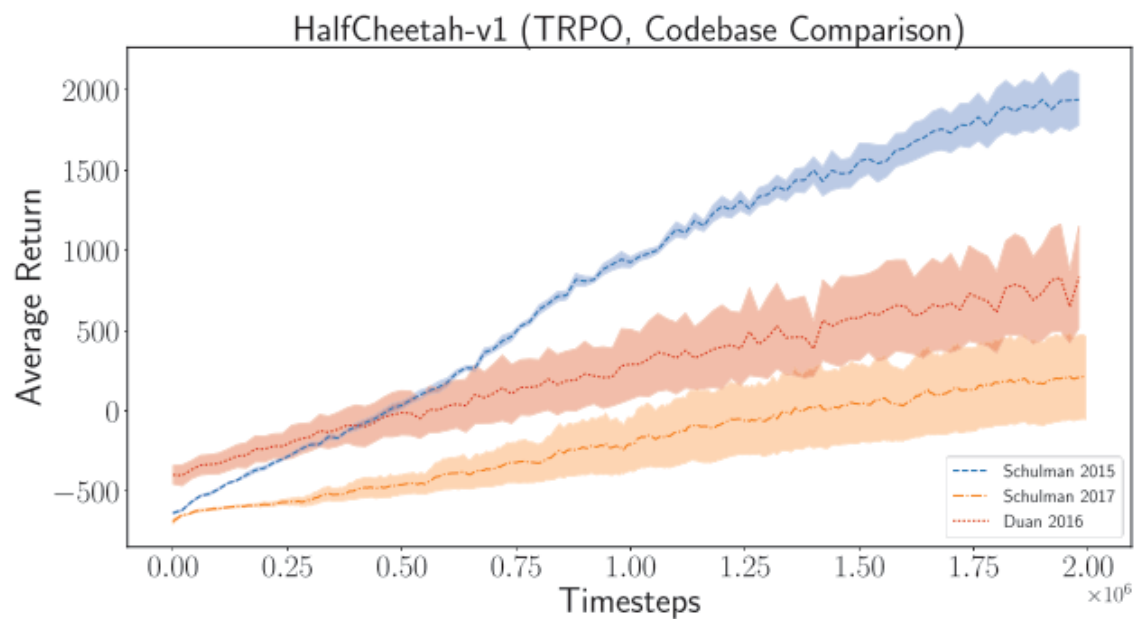


Figure 5: TRPO on HalfCheetah-v1 using the same hyperparameter configurations averaged over two sets of 5 different random seeds each. The average 2-sample  $t$ -test across entire training distribution resulted in  $t = -9.0916$ ,  $p = 0.0016$ .

# Codebases



# How to choose an RL Algorithm



# How to choose an RL Algorithm

(According to Sergey Levine)

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(According to Sergey Levine)



# How to choose an RL Algorithm

(According to Sergey Levine)



Sample  
Efficiency

# How to choose an RL Algorithm

(According to Sergey Levine)



Sample  
Efficiency

Ease of Use  
/ Stability

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Sample  
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← fewer samples



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Sample  
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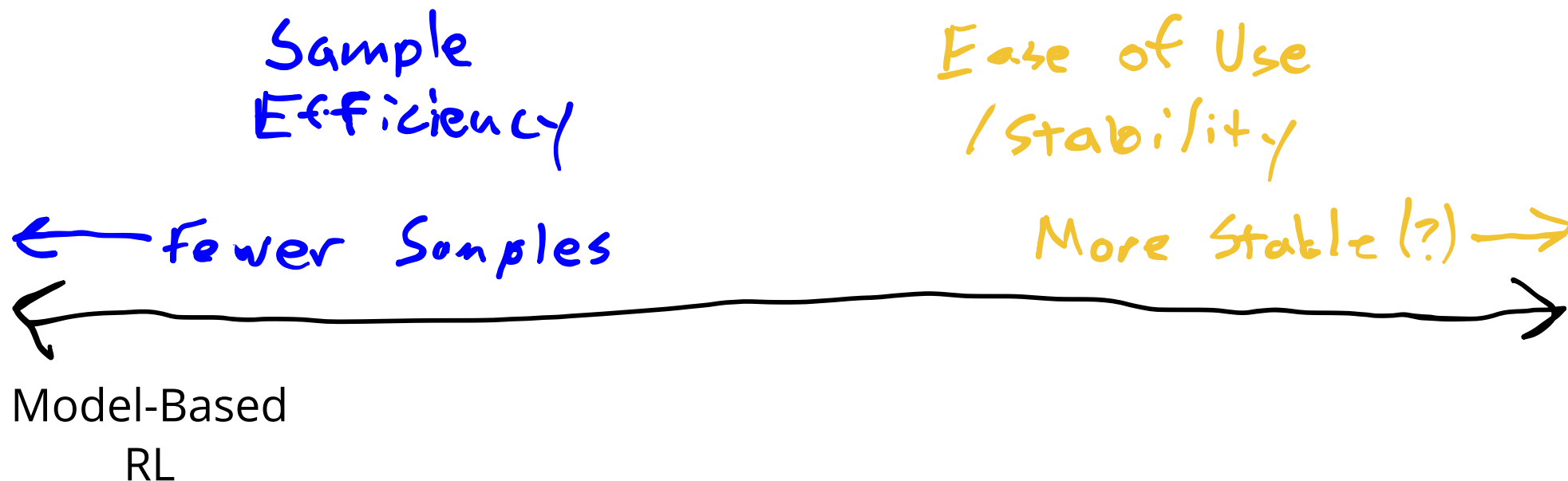
← fewer Samples

More Stable (?) →



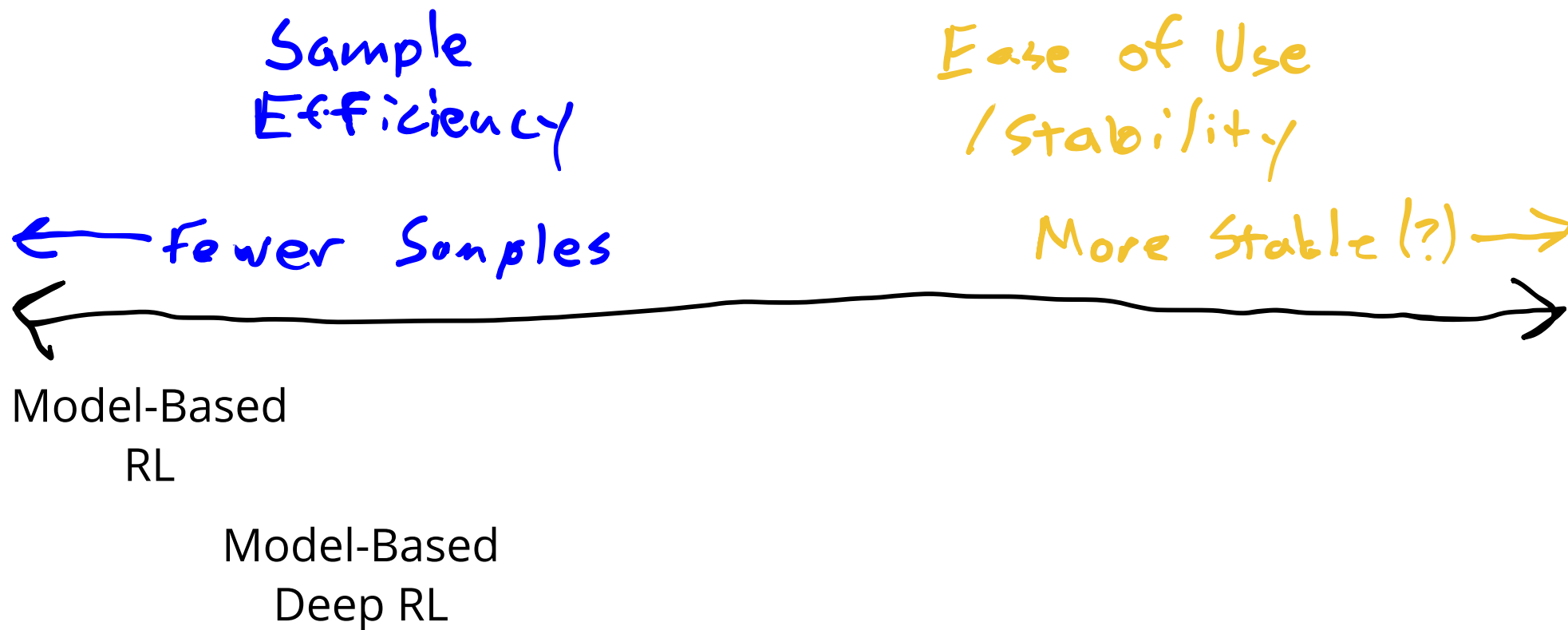
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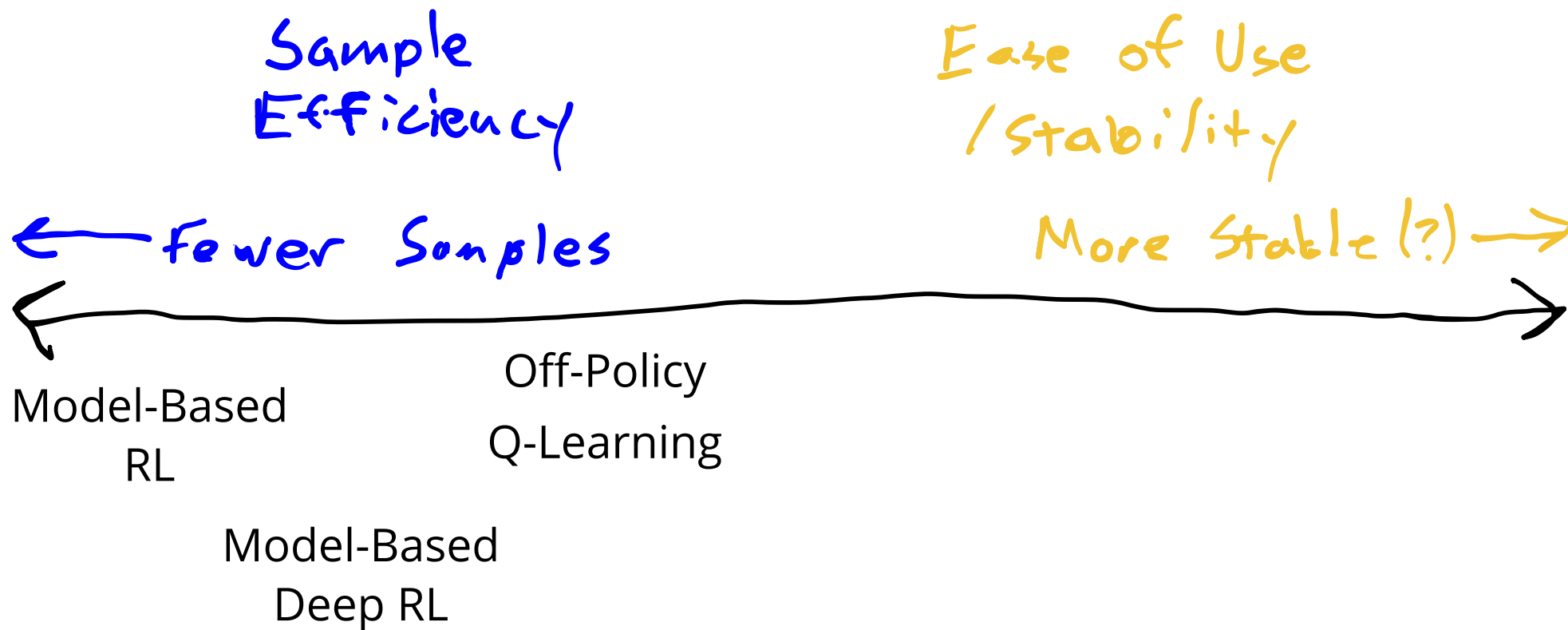
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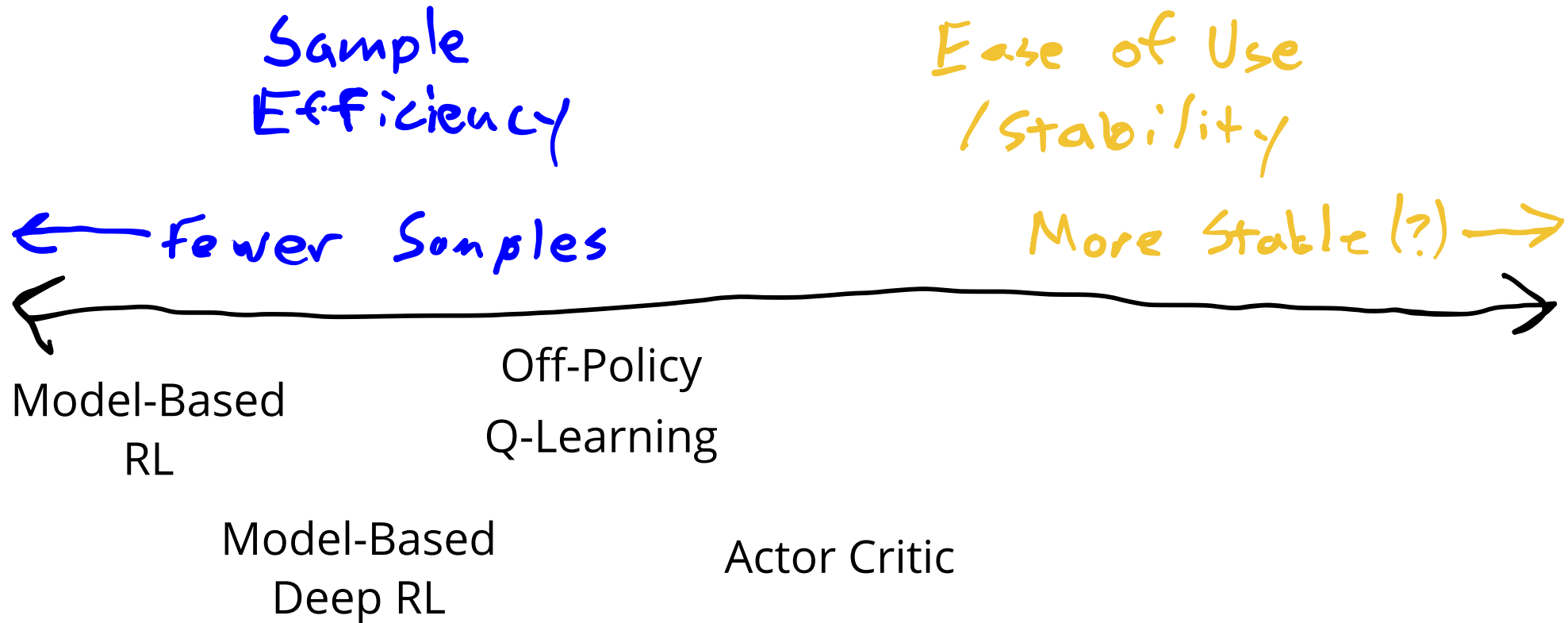
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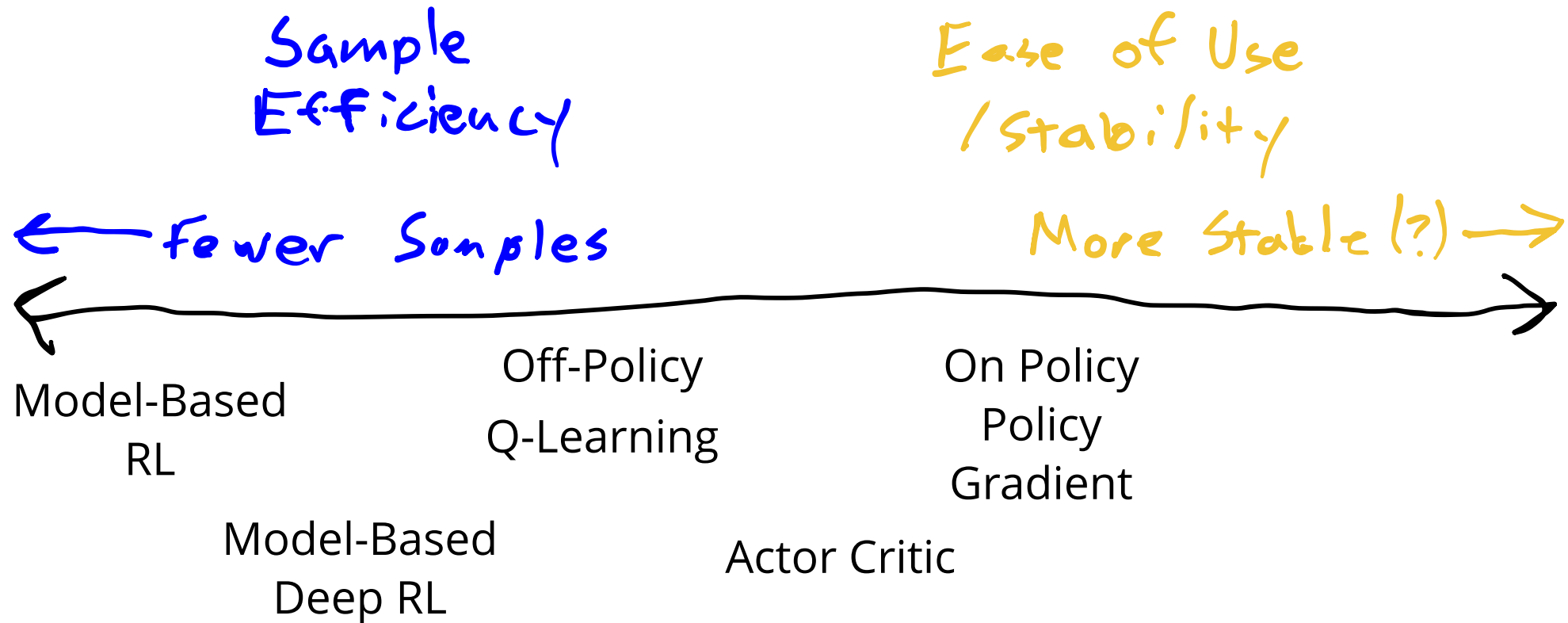
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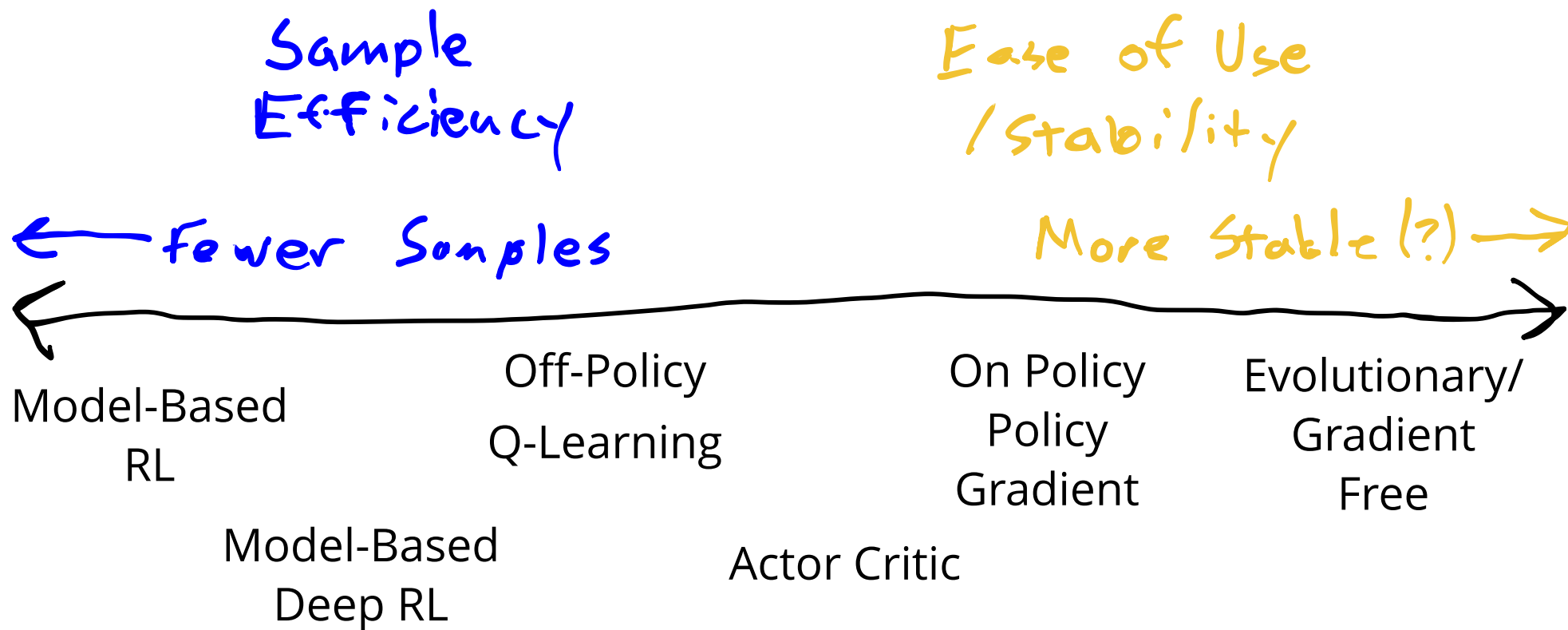
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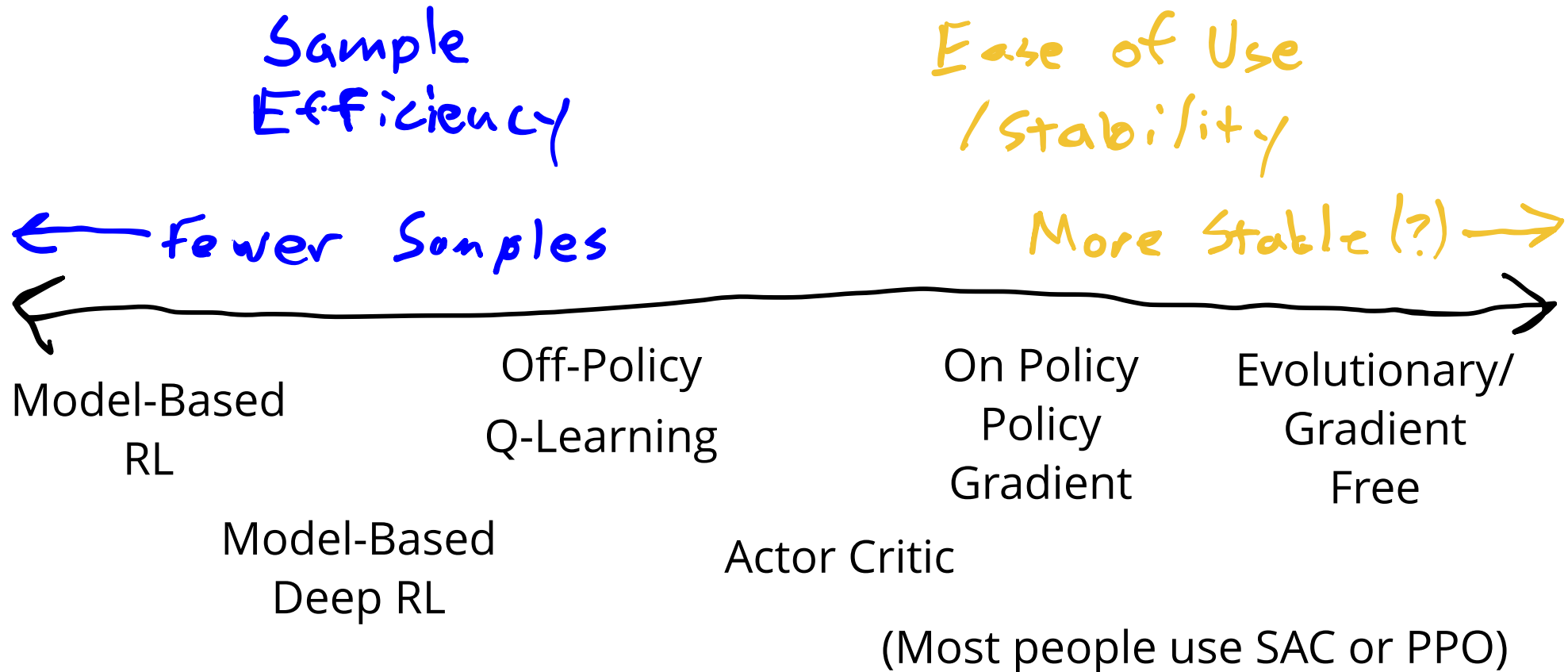
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# How to choose an RL Algorithm

(According to Sergey Levine)



# How to be successful with RL

- Always **start with a small problem that works** and scale up (keep verifying that it works with every change)
- Plot everything that you can think of (TensorBoard)
  - \*Losses\*
  - Policies
  - Value functions
  - Trajectories
  - (Average return) Learning curve
- Keep calm and lower your learning rate

